Sheltering Refugees in Cities: Crime, Public Services, and Voting

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April 18, 2026 - Click for the latest version

Abstract

An extensive literature documents that immigration flows frequently trigger native backlash, often reflected in heightened support for far-right parties. Yet, the mechanisms driving this response remain unclear, whether rooted in economic competition or cultural threat. Because these channels operate locally and through daily contact, identifying them requires granular data that are typically scarce. This paper examines how the reception of refugees affects local public education, crime, and ultimately voting behavior, exploiting detailed within-city data and plausibly exogenous variation from the quasi-random location of refugee shelters. Using a difference-in-differences strategy, I find that refugee shelters had no impact on crime or congestion in local public schools. However, they boosted natives' support for far-right candidates and reduced votes for the incumbent. Effects are largely driven by shelters hosting culturally diverse refugees comprising different indigenous ethnicities. Together, the results reveal that cultural perceptions can dominate economic channels in shaping local political responses to migration, and that aggregated data can miss essential nuances in locals' attitudes towards migrants.

Keywords: Refugees; Political Behavior; Public Services; Social Cohesion

JEL Codes: F22, O15, D72

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Acknowledgments: I am grateful to Giovanni Peri, Diana Moreira, and Monica Singhal for their invaluable support and feedback. I also thank Hani Mansour, Catalina Amuedo-Dorantes, Sascha Becker, Marianne Bitler, Jenna Stearns, Claudio Ferraz, Thierry Mayer, Delia Furtado, Yvonne Giesing, Simone Moriconi, Tony Payan, Panu Poutvaara, Youssef Ait Benasser, Arman Rezaee, Ashish Shenoy, Sandile Hlatshwayo, as well as the participants of the UC Davis Development Seminar, UC Davis Applied Micro Series, 2025 ASSA, 2025 European Economic Association, 2025 All-California Labor Economics Conference, 2025 LACEA, 2025 APPAM, 2025 SEA, 2024 CEMIR Junior Economist Workshop, 9th Understanding Voluntary and Forced Migration Conference, and 2024 Institute for Humane Studies Immigration Papers Workshop for providing insightful comments and suggestions.
Funding: UC Davis College of Letters & Sciences Dean's Graduate Summer Support Award and UC Davis Global Affairs Grad Grants.

1 Introduction

The number of refugees and people in need of international protection has more than tripled over the past decade, surpassing 41 million people in 2023. Low- and middle-income countries, despite facing limited state capacity and fiscal resources, host the majority (75%) of them - UNHCR Statistics.

At the same time that future displacement will likely intensify, an extensive literature documents that migration flows often trigger natives' discontent through increasing support for anti-migration, far-right, and populist candidates and parties.[1] Studies have emphasized economic channels, such as the labor market and welfare competition.[2] Others have focused on non-economic forces, such as media, propaganda, misinformation, and cultural differences.[3]

Much of this evidence relies on aggregated data, typically at the municipality or state level. While such approaches are informative about average public opinion, they may mask important heterogeneity in exposure. Natives living in directly affected areas are those who interact daily with migrants: they are potential employers and competitors in local labor markets, parents whose children share classrooms with refugee students, and neighbors whose streets and public spaces become sites of direct intercultural contact. As a result, their attitudes can shape migrants' social and economic interactions and undermine or reinforce the political sustainability of localized reception efforts, as suggested by the contact hypothesis and conflict theory - Allport, Clark, and Pettigrew (1954), Blalock et al. (1967), and Schlueter and Scheepers (2010).

This paper fills this gap by studying how the sudden and concentrated displacement of Venezuelan migrants into Brazil affected electoral and non-electoral outcomes in the receiving area. This large and sudden inflow of migrants was triggered by Venezuela's economic and political upheaval. Since 2014, nearly 8 million Venezuelans have emigrated, with the majority settling in neighboring countries. In 2018 alone, more than 150,000 Venezuelans crossed into Brazil through the border state of Roraima. In re-

[1] Mayda, Peri, and Steingress (2022) (U.S counties), Campo, Giunti, and Mendola (2021) (Italian municipalities), Dustmann, Vasiljeva, and Piil Damm (2019) (Danish municipalities), and Edo et al. (2019) (French departments).

[2] Becker, Fetzer, et al. (2016) (UK), Mayda, Peri, and Steingress (2022) (US), and Halla, Wagner, and Zweimüller (2017) (Austria).

[3] Campo, Giunti, and Mendola (2021) and Barone et al. (2016) (Italy), and Rozo and Vargas (2021) (Colombia).

sponse, the Brazilian government established urban shelters in the state capital, Boa Vista, and granted Venezuelans documentation and access to public services, welfare, and work permits.

To study the hyperlocal effects of urban refugee shelters, I assemble rich and granular data across several outcomes. Specifically, I examine detailed crime reports to capture externalities on safety and use comprehensive information on public schools to measure congestion of shared public service. I also examine voting behavior, a downstream measure of political attitudes and discontent, using geolocated pooling station election results.

I estimate the causal impact of refugee reception by leveraging different features of the Venezuelan displacement and reception policy in Brazil. First, migrant inflows and the government's response were highly concentrated in Roraima, with limited salience elsewhere in the country. Second, refugee assistance was implemented rapidly, leaving little time for local lobbying. Third, shelters were established in different areas of Roraima's capital by the military based on space availability and logistical constraints, rather than local political or economic conditions. I exploit this spatial and temporal variation in a difference-in-differences framework that compares areas near shelters with those farther away, before and after their establishment.

According to the results, refugee shelters reshaped within-city voting patterns. Proximity to shelters reduced support for the incumbent governor while increasing votes for far-right governor and presidential candidates by 2 to 4 percentage points. In contrast, I find no evidence that shelters influenced crime reports of robbery, homicide, and assaults. Similarly, although shelters raised the share of Venezuelan students in nearby schools, they had no discernible impact on classroom size, student–teacher ratios, or school infrastructure.

Importantly, the political effects were driven entirely by facilities hosting Indigenous Venezuelans. According to UN reports, Indigenous refugees were especially vulnerable, with higher adult illiteracy, lower vaccination rates, and lower levels of integration as measured by social security registration and work permit possession. Moreover, Indigenous refugee children had limited school enrollment, and more cases of child labor and homelessness were registered in those shelters' surroundings. This indicates that the salience of refugee presence and locals' exposure to poverty and vulnerability may underlie the political response. Therefore, ethnic-specific contact triggered political backlash even in the absence of worsening crime or public service congestion.

I complement the within-city analysis with a municipality-level synthetic control and synthetic Diff-in-Diff approaches, using non-affected municipalities in Brazil as the donor control pool. The results show that Boa Vista's overall far-right voting closely mirrored those of other non-affected comparable municipalities, indicating that the localized effects of shelters were not large enough to shift aggregate electoral outcomes. This finding assures that the within-city shelter variation captures the meaningful exposure to refugees and also shows how aggregate data can hide important nuances in natives' attitudes towards migrants.

The voting results speak to a large body of research in the field of political economy of migration, also documenting that inflows of migrants often increase support for populist, far-right, and anti-immigration candidates and parties.[4] [5] However, only a smaller and more recent set of studies uses more spatially granular data to study electoral behavior in response to refugee exposure. For instance, Fremerey, Hörnig, and Schaffner (2024) uses 1km grids across Germany, Endrich (2024) examines refugee housing centers in Hamburg, Pettrachin et al. (2023) focuses on refugee reception centers in Berlin, and Schmidt, Jacobsen, and Iglauer (2024) explores refugee accommodations across Germany. This paper offers different contributions. First, I complement the voting analysis with rich, granular data on public services and crime. Both are relevant potential mediators, particularly in developing country contexts characterized by higher crime prevalence and limited state capacity. In addition, the ethnic separation of refugee shelters allows me to isolate how the characteristics of the sheltered population shape political reactions.

Moreover, although developing countries host the majority of the world's displaced

[4]See Alesina and Tabellini (2024) for a review of the literature.

[5]Most existing studies examine electoral responses to migration across a range of different aggregated administrative units, including French cantons and municipalities (Edo et al. (2019) and Vertier, Viskanic, and Gamalerio (2023)), Italian municipalities (Barone et al. (2016) and Campo, Giunti, and Mendola (2021)), Spanish provinces (Mendez and Cutillas (2014)), regions in twelve European countries (Moriconi, Peri, and Turati (2022)), USA counties (Mayda, Peri, and Steingress (2022)), Austrian communities and municipalities (Halla, Wagner, and Zweimüller (2017) and Steinmayr (2021)), Swiss communities (Brunner and Kuhn (2018)), Greek islands (Dinas et al. (2019)), and Danish municipalities (Harmon (2018)).

population, evidence in these contexts remains limited.[6] This paper examines Brazil's reception policy, considered *"a global example"*, in a region where a history of violence, democratic erosion, and climate change pressures poses significant risks for future displacement.

This paper also contributes to the literature on political accountability, which studies how voters associate policies with policymakers.[7] Specifically, I examine the effect of shelters on support for the incumbent involved in the refugee reception policy. Considering that in most contexts migrants have limited to no voting rights, locals' political reactions and voting behavior can be consequential to shaping the implementation and maintenance of policies targeting immigrants. In fact, my results suggest that voters indeed react to refugee inflows once ethnic diversity and visible vulnerability enter their local neighborhoods.

More broadly, this study also expands the literature on the effects of refugee camps and shelters by looking at public education and crime. Prior work has focused primarily on protests, rents, earnings, employment, and consumption among local populations.[8] I complement this evidence by examining additional dimensions that capture key aspects of local communities' experiences of refugee arrivals.

The rest of the paper is organized as follows. First, I describe the inflow of Venezuelan refugees into Brazil, the reception policy, and the electoral system. The third section describes the data. Section 4 presents the empirical strategy and the tests for data quality and identification assumptions. In Section 5, I discuss the main results. I evaluate the differential effects between indigenous and non-indigenous shelters in Section 6. Finally, Section 7 concludes.

[6]Exceptions are Rozo and Vargas (2021) and Lebow et al. (2024), who look at Venezuelan migration in Colombia, and Ajzenman, Dominguez, and Undurraga (2022), who explores immigration in Chile.

[7]Ferraz and Finan (2008), for example, shows that Brazilian voters punished politicians when corruption was revealed in municipalities. Ajzenman and Durante (2023) shows that the infrastructure quality of polling stations in Argentina (usually public schools) signaled for the incumbent's education policies and shifted voters' decisions. See also Bobonis, Cámara Fuertes, and Schwabe (2016) and Weitz-Shapiro and Winters (2017).

[8]See Coniglio, Peragine, and Vurchio (2023), Batut and Schneider-Strawczynski (2022), Hennig (2021), Sanghi, Onder, and Vemuru (2016), Tumen (2016), Alix-Garcia, Walker, et al. (2018), and Alix-Garcia and Saah (2010).

2 Background

2.1 Time Variation: Venezuelan Refugee Inflow to Brazil

Venezuela's deep political and economic crisis led to a 65% decrease in its GDP between 2014 and 2019 and yearly inflation rates above 1000%.[9] Human Rights Watch reported constant violations of human rights, including the persecution of journalists and civil society organizations and the capture of the judiciary by the government. UNHCR estimates that 7.7 million citizens emigrated, more than 84% to other countries in Latin America and the Caribbean.[10]

Figure 1: Brazil-Venezuela Border and Roraima's Municipalities

Between January 2017 and April 2024, more than 1 million Venezuelans entered Brazil, and around half left to go to other South American countries.[11] Between 2018 and 2019 alone, almost 400,000 Venezuelan border crossings were registered in the border state of Roraima, representing more than 80% of the national inflow - see Figure 2.[12] After a sharp decline in 2020 and 2021, when the border was closed due to the COVID-19 pandemic, entries rebounded in 2023 and 2024, suggesting that migration inflow pressures remain in the region. [13]

[9] IMF statistics.

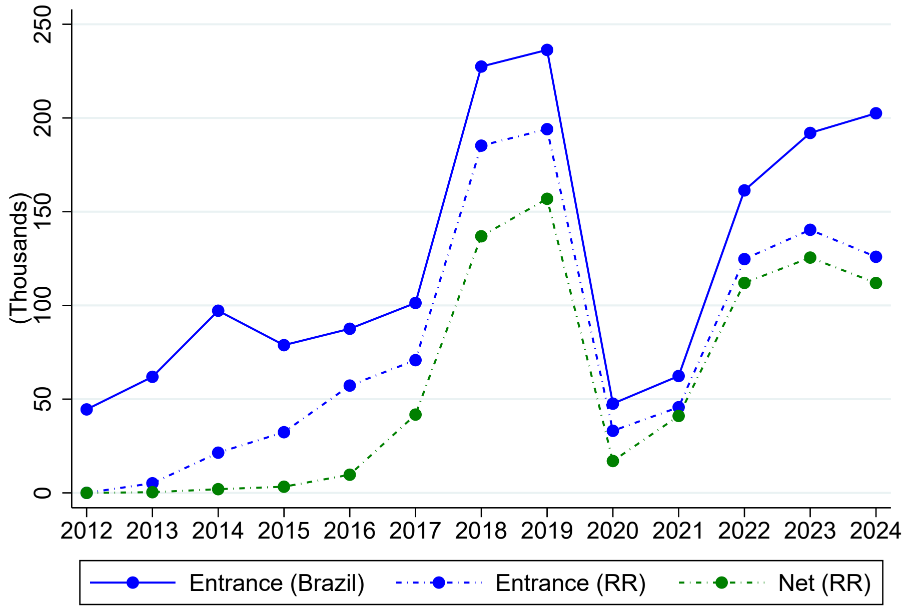
[10] See R4V Platform for statistics by hosting country.

[11] Source: Ministry of Justice and Public Security report on Venezuelan Migration for April 2024.

[12] The rise in net inflows to Roraima suggests that Venezuelan inflow was not limited to short-term cross-border trips for goods and supplies, for example.

[13] See Figures 29 and 30 in the Appendix A for more details about the gender and age composition of the inflow.

Figure 2: Venezuelan Migration Inflow to Brazil and Roraima



Source: Sistema de Tráfego Internacional (STI).

After crossing the border at Pacaraima, migrants typically travel to Boa Vista, Roraima's capital and largest city, with a population of nearly 400,000 in 2017, where they can access transportation and infrastructure to reach other parts of the country - see Figure 1. Although Boa Vista serves as a transit hub, it also hosts a large and highly visible refugee population. According to a survey conducted by the Boa Vista government in June 2018, 25,000 refugees were living in the city (7.5% of its population), and around 10% were homeless.[14]. Boa Vista became a major national point of refugee concentration and visibility, given that the foreign-born population in Brazil accounted for less than 1% of the total population in 2017.

I combine UNHCR shelter reports with data from the Brazilian state-level household survey (PNAD) to compare the demographic composition of Roraima's population and sheltered refugees. The refugee population is substantially younger, with a higher concentration of children aged 11 or below and a lower share of individuals aged 60 or above. This difference in age structure highlights the potential for increased demand on public education services.[15]

[14]See newspaper article

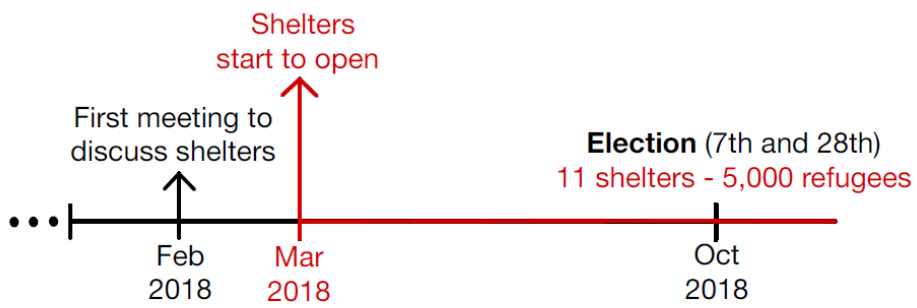
[15]PNAD data are only available at the state level and don't allow us to separate foreign and Brazilian individuals. Therefore, if anything, the differences between the two populations are underestimated. For more details, see Figure 31, 32, and 33 in the Appendix.

2.2 Spatial Variation: Reception Policy

In Brazil, immigrants and refugees, regardless of their legal status, have free movement within the country and can access public schools and the free, universal national health care system. Once documented, they can access the formal labor market and welfare programs, such as cash transfers. Venezuelan migrants were regularized through refugee status or residency permits, with documentation rates close to 100%.[16]

The Brazilian Federal Government launched the *"Operação Acolhida"* (*Reception Operation*) in February 2018 to deal with the increasing number of refugees crossing Roraima's border. The operation consists of a humanitarian task force coordinated by the federal, state, and local governments with UN agencies, international and civil society organizations, and private entities. Different reception, accommodation, regularization, sanitary inspection, and immunization structures were set up at the border and in Boa Vista. The Operation consisted of three main foundations: border planning, dispersal policy, and reception.[17]

Figure 3: Timeline



Shelters started to be open in March, they provided food and protection for documented refugees. Teams of NGOs, UN, and government workers offered health services, portuguese classes, and informal education activities for children. Some shelters consisted of the "Refugee Housing Units" model of the UNHCR, while others had tents and overlays provided by the Brazilian army - see Figure 4. The shelters were jointly

[16]This reception differs from some European countries, where government displacement policies place all arriving refugees in specific municipalities, and refugees can face limited access to the labor market immediately after arrival. For example, asylum seekers are obligated to stay in reception centers during their initial asylum proceedings in Germany and throughout their refugee status determination process in Denmark - see Ginn et al. (2022).

[17]Since April 2018, more than 140,000 Venezuelans have participated in the voluntary dispersal policy and moved to more than 750 Brazilian municipalities. For updated statistics about the dispersal policy access: Dispersal Strategy Statistics Platform.

managed by the Brazilian army, NGOs, UNHCR, and state and municipality governments. The bathrooms were shared, and some shelters didn't have a dining area. Entry was permitted until 10 pm, and identification cards were used to control entry.[18] In October 2018 (the election month), 5,000 refugees lived in one of the 11 shelters in Boa Vista. From the opening, shelters were at full capacity (some above it), the smallest one hosted 279 Venezuelans, and the biggest more than 650 refugees.[19]

Figure 4: Shelters' Photos

(a) Inside Photos

(a.1) Tancredo Neves Shelter

(a.2) Rondon 1 Shelter

(b) "Operação Acolhida" logo and shelters' name on outside signs

(b.1) Jardim Floresta Shelter

(b.2) Santa Teresa Shelter

2.3 Elections

Voting is mandatory and restricted to Brazilian citizens. The naturalization of individuals without specific family ties with Brazilians can take up to 180 days and requires

[18]For more details about the shelters' logistics and the discussion behind the militarization of the reception policy, see Machado and Vasconcelos (2022).

[19]See Table 18 in the Appendix B for shelter-specific statistics.

a minimum number of years living in the country (4 years in most cases), besides proof of Portuguese proficiency. Therefore, Venezuelan refugees are very unlikely to be voting.[20]

Elections take place every two years, in even years, alternating between municipal and general elections. A second round for executive positions is held if no candidate obtains a majority. On October 7th, 2018, more than 150,000 registered voters in the Boa Vista voted for: President, State Governor, National Congress (8 vacancies), Senators (2 vacancies), and State Congress (24 vacancies). Since no candidate for President and Governor reached 50% or more of the valid votes, a second round was held on October 28th.

Presidential Election Candidates

"Operação Acolhida" was launched during Michel Temer's administration.[21] He decided not to run for reelection; therefore, there was no incumbent candidate in the 2018 presidential election.[22] The main national left party, the Workers Party (PT), launched Fernando Haddad, who lost the 2018 election to Jair Bolsonaro (55% of the votes). Jair was a federal deputy for the Rio de Janeiro State for 27 years (6 consecutive reelections). He was known for his conservative and controversial statements.

"Refugees arriving in Brazil are the scum of the world."

Bolsonaro (2015)

The Venezuelan migration crisis was not a major part of the national presidential debate. However, Haddad (PT) and Bolsonaro (PSL) had considerably different views about immigrants. The 2018 Bolsonaro government program doesn't mention immigrants or refugees directly. Contrastingly, Haddad's program explicitly aimed to promote refugees' and immigrants' rights and refers to them as a target population for public policies.

[20]Information about the number of naturalized citizens among the voters is unavailable. However, naturalized citizens represent less than 0.1% of the country's total population according to the 2022 Census.

[21]For a complete timeline of recent Brazilian Presidents, see Figure 35 in Appendix C.

[22]His party launched the finance minister as a candidate, who got less than 1.3% of the valid votes nationally.

Haddad's (PT) Presidential Government Program (2018):

"The Government will promote the rights of migrants through a National Migration Policy and will broadly recognize the rights of refugees."

"Health improving actions will be implemented for women, ..., immigrants, refugees, ..., and people from the forests."

PT had a candidate in all presidential elections in the data (2006 to 2022). However, for some election years before 2018, Bolsonaro's Party (PSL) didn't launch a candidate, so I will use the voting for the candidate it supported. See Table 20 in Appendix **??** for a complete description of the parties used in each presidential election.

Governor Election Candidates

Both media coverage and online search behavior indicate that the Venezuelan migration crisis was highly salient in Roraima during the 2018 election period. Google Trends data show that search intensity for terms such as "Venezuelans," "refugees," and "shelters" far exceeding those observed in other Brazilian states - see Figure 5 and Table 1.

Figure 5: National Newspaper Headlines Covering Roraima's 2018 Election

ELEIÇÕES 2018 · VENEZUELA

Crise migratória vira principal assunto da eleição em Roraima

Candidatos se unem em discurso contra refugiados, e governadora diz que fronteira aberta afeta popularidade

Na eleição de Roraima, o que importa mesmo é a Venezuela

Candidatos ao governo de Roraima debatem sobre a entrada de imigrantes venezuelanos pelas fronteiras

Translation: *"Migration crisis becomes the main issue of the election in Roraima"* and *"In Roraima's election, what really matters is Venezuela"*

Table 1: Google Trends Search Intensity Index for 2018

	"Venezuelans"	"Venezuela Crisis"	"Refugees"	"Shelters"	"Operacao Acolhida"
Roraima	100	100	100	100	100
2nd State	10	27	41	65	8
3rd State	4	24	38	46	1

Notes: Google Trends search intensity index ranges from 0 to 100, where 100 represents the maximum search interest (as a share of total searches) among all Brazilian states.

From 2014 to 2018, Suely Campos (of PP, an important national right-wing party) was Roraima's Governor. She won the 2014 election with 54.9% of the votes and unsuccessfully ran for reelection in 2018 (less than 12%). Her administration participated directly in the *Reception Operation* efforts. The state received extra funds for social and health services and, together with the federal government, created different commissions addressing topics related to the refugee flow. The state government also directly managed two shelters and participated in interventions targeting the sheltered population (such as STD testing, distribution of condoms, vaccine campaigns, and nutrition surveillance).

Antônio Denarium - 2018 Roraima's Elected Governor:

> *"Together with refugees, drug dealers, and criminals are entering; one country, Venezuela, does not fit inside Roraima."*
>
> *"... all these NGOs that are here should go to Venezuela and serve these people there, preventing them from entering Brazil."*
>
> *"...(we want to) restrict the entry of Venezuelans by requiring a passport, a criminal record certificate, and a vaccination certificate, which is also very important.*

Antônio Denarium was elected governor in 2018 with 53.34% of the votes. He shared the same party (PSL) as Bolsonaro, who visited Roraima and joined him in campaign events. Both Suely, and Denarium defended some form of border restriction.[23]. I therefore interpret voting for Suely as a measure of political accountability—capturing whether voters reward or punish the incumbent for her role in the refugee response—and voting for Denarium as reflecting support for a challenger aligned with the main Far-Right national candidate. Similarly to the presidential election, I will complete the years with no candidate from such parties with the voting for the party they support.[24]

[23] Roraima's Governor Candidates Interview.

[24] For more details on political context, see Appendix C and Table 19.

3 Data

3.1 Shelters and Refugees

UNHCR produced a summary of the *Reception Policy* with a timeline of shelter openings and a description of services and programs targeting Venezuelans. Additionally, shelter-specific monthly reports published in 2018 contain shelters' locations, total capacity, and their sheltered refugees' socioeconomic and demographic information. Government meeting minutes available at the *Receptio Policy* website were used for shelters and months not covered by the UNHCR reports.

3.2 Election

Data for the 2006, 2010, 2014, 2018, and 2022 election results is provided by the Superior Electoral Court (TSE). It contains the number of votes for each candidate in each section (room) in each polling station (building). Additionally, from the 2010 election onwards, the characteristics (age, sex, marital status, and education) of the registered voters are also provided at the section level.[25]

I use data on the geographic coordinates of polling stations from F. Daniel Hidalgo. It leverages different administrative datasets to fuzzy string match the address and the polling station name (usually the name of the building). The coordinates come from TSE data and other administrative datasets (such as schools' geographic location from the Education Ministry).[26]

3.3 Crime

Crime data for Boa Vista were provided by Roraima's State Police Department (SESP). The dataset includes all reported and attempted incidents of homicide, robbery, assault, rape, and felony murder (robbery resulting in death). Incidents are recorded at the address level and include the exact date (day, month, and year) of occurrence. The data cover the period from January 2016 to December 2022.

[25] The marital status information contains a considerable amount of missing; therefore, only voters' education, gender, and age were used.

[26] See Appendix D for the details on how this data was used and the procedures taken to confirm each polling station's latitude and longitude.

First, I used the list of all streets, avenues, and alleys in Boa Vista from the Brazilian Postal Service to string match each crime address, correcting it for typos, spelling, and accentuation errors. To geo-locate the cleaned addresses, I use the OpenCage Geocoding API that uses different open datasets (like OpenStreetMap) to find the best possible latitude and longitude match. Around 80% of the crime data was successfully geolocated. I excluded addresses in the rural area of the municipality and dropped observations without latitude and longitude. The final data includes 24,063 reports, 59% are robbery, 35% are assault or battery - see Table 2.

Table 2: Crime Reports Statistics

Crime Type	**Number of Reports (% Total)**	**% Classified as "Attempted"**
Robbery	14,117 (58.7%)	2%
Assault/Battery	8,339 (34.6%)	2%
Homicides/Murder	1,096 (4.5%)	47%
Rape/Sexual Assault	436 (1.8%)	18%
Felony Murder (robbery resulting in death)	75 (0.3%)	33%
Total	**24,063**	**5%**

Source: NEAC/DENARC/PCRR reports from January 2016 to December 2022.

3.4 Education

The administrative Ministry of Education's yearly school census covers all public schools in Boa Vista from 2010 to 2024. It provides information on location, staff, infrastructure, and students' age, gender, and nationality. I exclude rural and private schools from the sample, given that this paper focuses on the urban area of the municipality, and private schools accounted for less than 1% of Venezuelan enrollment. In 2017, the year before shelters were set up, the city had 118 public urban schools with an average of 495 students each, and 26 students per classroom - see Table 3.

Table 3: Public Schools Statistics - Boa Vista (2017)

	N	mean	sd	min	max
Share of Venezuelans (2019)	109	12.05	10.14	0	60.14
Number of Students	118	494.8	290.4	38	1,580
Number of Classrooms	118	18.53	9.935	2	57
Auditorium (Dummy)	118	0.144	0.353	0	1
Library or Reading Room (Dummy)	118	0.822	0.384	0	1
Science Lab (Dummy)	118	0.212	0.410	0	1
Students per Classroom	118	25.71	3.514	12	34.35
Students per Teacher	118	18.97	4.898	7.053	32.83
Distance Closest Shelter	118	1.684	1.054	0.149	4.518
Average Distance to Shelters	118	4.482	1.289	2.820	7.960

Source: 2017 and 2019 *School Census*.

4 Empirical Strategy

4.1 Empirical Strategy: Voting

Given the different aggregation options allowed by the detailed voting data, I first describe the chosen unit of observation for the main specifications.

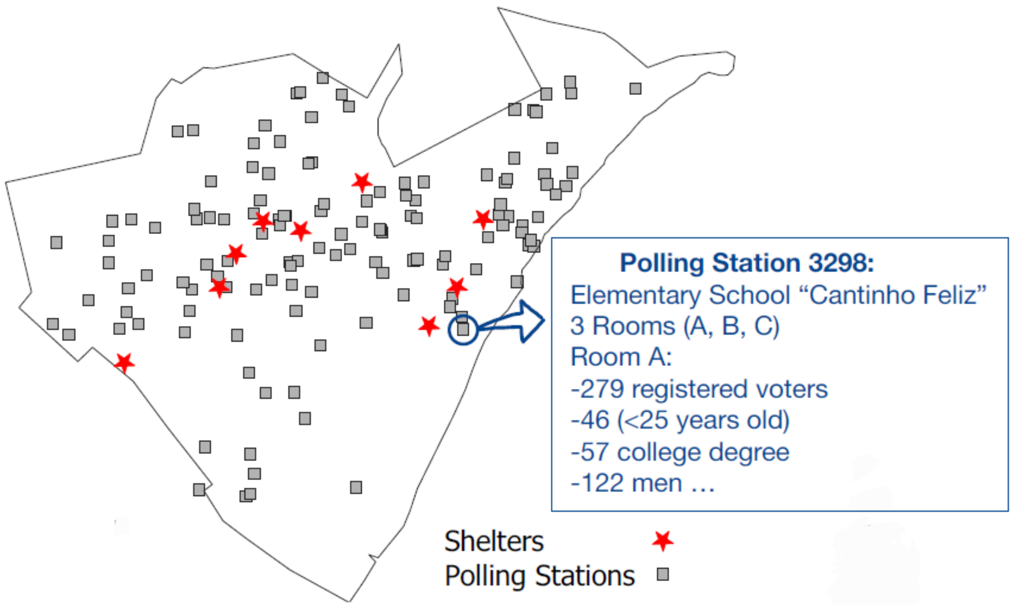
Brazil's electoral system does not rely on voting districts to allocate voters; instead, voter assignment operates across two levels. First, voters are assigned to a polling station (i.e., a building, usually a public school). Then, they are separated into different sections (i.e., rooms) within that building. The Brazilian Electoral Code describes the criteria behind those assignments:

> *§ 1º (...) (Polling Station) will be located within the judicial or administrative district of your residence and the closest to it, considering the distance and means of transport.*

Moreover, according to § 3º, the voter will be permanently linked to the room number indicated in their voter's ID. If voters move to another municipality, they must go to the office and update the polling station. If voters move within the same municipality to a different neighborhood, they can (not mandatory) update their polling station to one closer to their new residence address.

Therefore, the assignment of voters to polling stations and sections presents two interesting features for the purpose of this study. First, the location of polling stations can work as a proxy for the area of residency. Second, there is a certain inertia once you are assigned to a room (it is unlikely to be voting at different places or even rooms in each election). Given these electoral logistics features, I explore a room-level panel as the main dataset - see Figure 6 for an example of how the data looks.

Figure 6: Election Data on a Map



To obtain the causal impact of the shelters on the electoral outcomes, I estimate a Diff-in-Diff with the following TWFE specification:

$$Y_{ijt} = \beta \text{ Treated}_j \times Post_t + \gamma_i + \mu_t + Controls_{ijt} + \epsilon_{ijt} \tag{1}$$

Y_{ijt} is the voting outcome of the section "i" in polling station "j" in the electoral year "t". Treated$_j$ is a dummy variable indicating whether polling station "j" is less than 1 kilometer away from the closest Venezuelan refugee shelter. μ_t is the year fixed effects and γ_i is the section fixed effect. The 2006, 2010, and 2014 elections are the pre-treatment periods, and 2018 and 2022 are the post-treatment periods. Standard errors are clustered at the polling station level, and neighborhood-level clustering was also explored to assess robustness to underlying spatial correlations. $Controls_{ijt}$ includes 23 different demographic variables (interaction of education, gender, and age categories) of section "i" registered voters, and an interaction of time dummies and the distance of polling station "j" and the city downtown.[27]

Table 4: Voting Data Descriptive Statistics (2006-2022)

	N	mean	sd	min	max
Polling Station Constant Variables:					
Distance (Km) to the closest shelter	238	1.480	1.056	0.168	5.014
Average Distance (Km) to all shelters	238	4.689	1.282	3.161	9.188
Distance (Km) to Boa-Vista center/downtown	238	4.731	3.165	0.212	10.18
Treatment Dummy (1 km)	238	0.340	0.475	0	1
Section-Level Variables (2006-22):					
Number of Registered Voters	1,190	326.3	65.46	75	444
Turnout Rate 1st Round	1,190	85.40	4.009	70.19	95.57
Turnout Rate 2nd Round	1,190	81.82	4.614	57.47	95.07
Section-Level Variables (2014):					
Share Illiterate	714	1.258	1.387	0	7.407
Share with some college	714	28.66	18.44	0	82.78
Share Less than High-School	714	39.67	17.37	3	91.49
Share < 25 Years Old	714	16.87	8.911	0	58.11
Share < 40 Years Old	714	42.48	14.40	3.333	77.78
Share > 65 Years Old	714	7.821	5.473	0	40.79
Share Men	714	47.32	7.523	16.67	78.95
Share of less than High-school degree Men	714	20.79	10.39	1	70.53
Share of less than High-school degree Women	714	18.88	9.112	0	55.66

Source: TSE.

The data includes 911 sections (in 238 polling stations), each with 330 voters on average, 33% are located in treated polling stations, and 28% of the sections are balanced (operate every election year in my data). Sections can be destroyed or created during this period for different reasons, such as changes in the voters' population size or logistical reasons, such as building renovations.

Considering that covariates can bias the TWFE even in a non-staggered design with two time periods (Sant'Anna and Zhao (2020). Callaway and Sant'Anna (2021)), I also estimate a Doubly Robust Diff-in-Diff (DRDiD) using the 2014 voters' characteristics covariates. Additionally, I estimate a Matching Diff-in-Diff that first uses 2014 covariates to match control units to treated units, then calculates a conventional

[27] Geographic controls based on distance to downtown are also added in Hennig (2021). Moreover, Barsanetti and Pinho Neto (2025), shows that distance to city center explain the distribution of overall infrastructure in Brazilian cities.

Diff-in-Diff.[28]

Data Quality and Identification Assumptions

Next, I will discuss and test the identification assumptions and threats to the causal interpretation of "β".

Exogenous Location of Shelters

According to the minutes of the reception operations meeting, the Defense Ministry was responsible for selecting locations and visiting available land. Shelters were either established in areas around the Federal Police headquarters or in empty areas and buildings, such as public gymnasiums. The available documents do not provide detailed criteria or describe specific strategies guiding location choices. However, the central role of the military, rather than local governments, in determining these locations provides an institutional basis for parallel trends in electoral outcomes.

Table 5: Balance Table on Pre-Migration Variables

		Treatment			Control		
	n	mean	sd	n	mean	sd	Diff
Section Constant Variables:							
Distance (Km) to the closest shelter	81	0.59	0.25	157	1.94	1.02	-1.354***
Average Distance (Km) to all shelters	81	3.87	0.52	157	5.11	1.35	-1.248***
Distance (Km) to Boa-Vista center/downtown	81	5.62	2.85	157	4.27	3.23	1.352***
Section Variables (2014):							
Share Men	81	47.27	3.26	157	47.09	3.89	0.178
Share <25 Years Old	81	19.82	6.02	157	18.20	7.54	1.620*
Share <40 Years Old	81	48.94	11.19	157	46.80	13.82	2.143
Share >65 Years Old	81	5.50	3.05	157	6.05	3.91	-0.544
Share Illiterate	81	1.75	1.08	157	1.21	1.37	0.538***
Share Less than High-School	81	46.01	13.01	157	34.46	17.02	11.552***
Share with some college	81	20.44	12.20	157	33.69	19.49	-13.252***
Share of less than High-school degree Men	81	24.24	7.28	157	17.82	8.74	6.418***
Share of less than High-school degree Women	81	21.77	6.53	157	16.64	8.73	5.134***

Notes: Includes only balanced sections (main sample). *Source*: TSE.

One could also argue that locals may have engaged in lobbying to prevent shelters from being established in certain areas. If lobby movements existed (no media found

[28] The DRDiD is a combination of outcome regression and IPW (propensity score model) and was implemented using the command *csdid* in Stata. For the Matching DiD (MDiD), I use the command "diff" in Stata that runs a kernel-based propensity score matching.

on it) and were connected to locals' negative attitudes towards migrants, this would attenuate the estimated effects on far-right and incumbent performance. Moreover, "Operação Acolhida" was considered an emergency task force, and shelters began opening a month after the first operation meeting. Consequently, lobby organizations would have had a considerably limited time to organize.

Table 5 reports the differences between control and treated units on registered voters' demographics pre-shelter (2014). Treated sections have fewer educated voters and are farther from the city's downtown. As long as those differences are attributable only to differential levels of political preferences and not to trends, the DiD assumptions hold. To test this formally, I estimate an event-study version of equation (1) to verify whether there is any statistically significant pre-treatment effect of the shelters.

Quality of Polling Station Outcomes as Measures of Nearby Residents' Preferences

Voting outcomes at the polling station level should reflect the political preferences of residents living in the surrounding area. As discussed in Section **??**, this is partially ensured by the Brazilian Electoral Code. However, individuals who move within the same municipality are not required to update their polling station, which may weaken the spatial link between voting location and residence.

In 2013, the introduction of biometric registration in Boa Vista required voters to update their information, including fingerprint records, creating an opportunity to change polling stations if they were located far from their residence.[29] This reform likely strengthened the correlation between voting location and place of residence. Although voter address data are not publicly available to directly test this relationship, I provide indirect evidence by showing substantial stability over time in the demographic composition of polling station voters (education, age, and gender) after 2013 - see Appendix E.

[29] According to the TSE: "Some voter registration data are confidential (membership, address, telephone, date of birth, biometric data, among others) and must be updated whenever necessary, such as in cases where the voter must change personal data, *register fingerprints*, request transfer, etc."

Spillover Effects

Potential treatment-to-control leakage, likely to occur to happen in this within-city setting, would violate the SUTVA assumptions of the Diff-in-Diff and attenuate my estimates.

To address this concern, I restrict the control group to "likely untreated" polling stations, defined as those located farther than the median distance from the nearest shelter. I additionally estimate Equation (1) using three concentric distance rings around shelters: 0–1 km, 1–2 km, and 2–3 km. Finally, I also estimate a version of equation (1) using the distance to the closest shelter as a continuous treatment. ZDistance_j is the z score among all polling stations of the distance in kilometers between polling station "j" and the closest refugee shelter. This allows for a more flexible shelter effect across the Boa Vista urban area.

$$Y_{ijt} = \beta \ \mathit{Z(Distance)}_j \times \mathit{Post}_t + \gamma_i + \mu_t + \mathit{Controls}_{ijt} + \nu_{ijt} \tag{2}$$

Locals' endogenous migration and assignment to polling stations

Finally, we also assume that the voters have no compositional change due to treatment assignment. In other words, Brazilians (especially the most conservative/anti-migration ones) didn't move in response to shelters. This would represent a compositional change in our sample, leading to misleading zero or even wrong sign results.

Electoral logistics further mitigate this concern. Voters had until May 9, 2018, to update their polling station, while 8 out of the 11 shelters opened after March 2018, leaving limited time for voters to adjust their registration in response to shelter locations. As a result, even if locals changed their residences, responding to the shelters' location, we would still likely capture their political preferences in their original polling station. Nonetheless, the possibility of moving gives the estimates an ITT interpretation.

To formally test if treatment affected voters' characteristics (a sign of voters' compositional change), I estimate equations (1) and (2) using those voters' education, age, and gender characteristics as outcomes. According to the results, the shelter's location had a null effect on different voters' characteristics - see Table 6.

Table 6: Diff-in-Diff Estimates - Demographic Control Variables as Outcome

Outcomes	$\boldsymbol{Treated_j} \times \boldsymbol{Post_t}$	**R2**	$\frac{1}{\boldsymbol{Distance_j}} \times \boldsymbol{Post_t}$	**R2**
% Men	-0.876 (0.674)	0.002	0.118 (0.330)	0.001
% Illiterate	-0.135 (0.118)	0.025	0.023 (0.035)	0.023
% Less than High School	0.495 (0.894)	0.029	-0.503 (0.349)	0.031
% Some College	0.302 (0.715)	0.008	0.404 (0.257)	0.009
% 16-17 Years Old	-0.331 (0.587)	0.042	-0.349* (0.181)	0.050
% <25 Years Old	1.317 (1.355)	0.087	-0.924 (0.761)	0.090
% >65 Years Old	0.083 (0.361)	0.336	-0.022 (0.155)	0.336
% Men Less than High School	-0.099 (0.685)	0.015	-0.610*** (0.215)	0.020
% Working-Age Men	-0.504 (0.786)	0.022	0.375 (0.311)	0.022
% Women Less than High School	0.594 (0.644)	0.009	0.108 (0.213)	0.008
% Working-Age Women	0.732 (0.771)	0.035	0.082 (0.259)	0.034

Notes: standard errors clustered at the polling station level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Differential Electoral Logistics

Another concern for the causal interpretation of the results would be that election logistics differed in sections closer to shelters. For example, polling stations closer to shelters could have been manipulated to make voting more difficult by increasing the number of registered voters or reducing the number of rooms. Following the same strategy used to investigate voter compositional change, I estimated a version of equations (1) and (2) using election logistics variables, both at the section and polling station levels, as the outcome.

According to the results, there was no differential logistic capacity between treated and control units - see Table 7. Therefore, it is unlikely that the election organization explains the results.

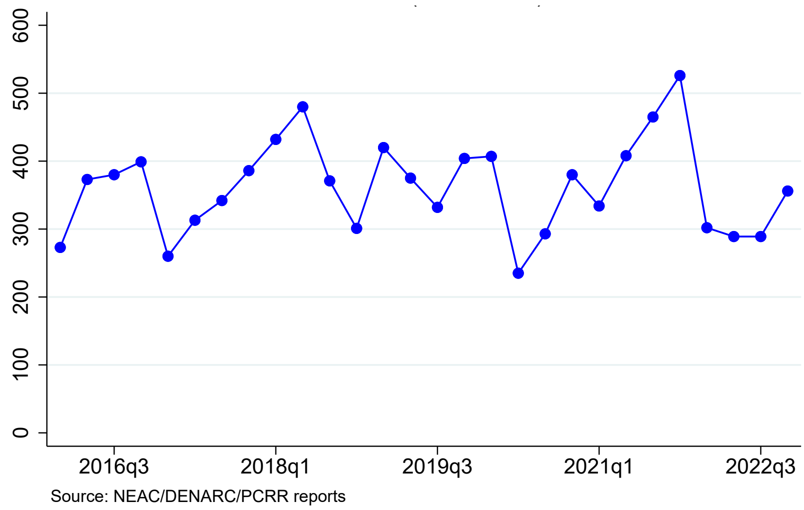
Table 7: DiD Results - Election Logistics Variables as Outcome

Outcomes	$\boldsymbol{Treated_j \times Post_t}$	**R2**	$\boldsymbol{\frac{1}{Distance_j} \times Post_t}$	**R2**
Polling Station Level:				
Number of Sections	0.069 (0.160)	0.308	-0.087 (0.199)	0.308
Number of Registered Voters	-14.628 (70.093)	0.167	-32.307 (75.469)	0.168
Average Section Size	-9.938 (9.391)	0.135	-2.453 (3.890)	0.134
Size Biggest Section	-6.321 (9.709)	0.176	-0.444 (4.163)	0.176
Size Smallest Section	-14.820 (15.699)	0.073	-5.168 (7.343)	0.073
Not Operating in year t	-0.009 (0.038)	0.129	0.023 (0.026)	0.132
Section Level:				
Number of Registered Voters	3.772 [11.785]	0.335	2.088 [2.869]	0.335
Not Operating in year t	0.011 [0.057]	0.315	-0.025 [0.034]	0.316

Notes: robust standard errors in parentheses and standard errors clustered at the polling station level in brackets. *** p<0.01, ** p<0.05, * p<0.1.

4.2 Empirical Strategy: Crime

Figure 7: Quarterly Number of Crime Reports - Boa Vista



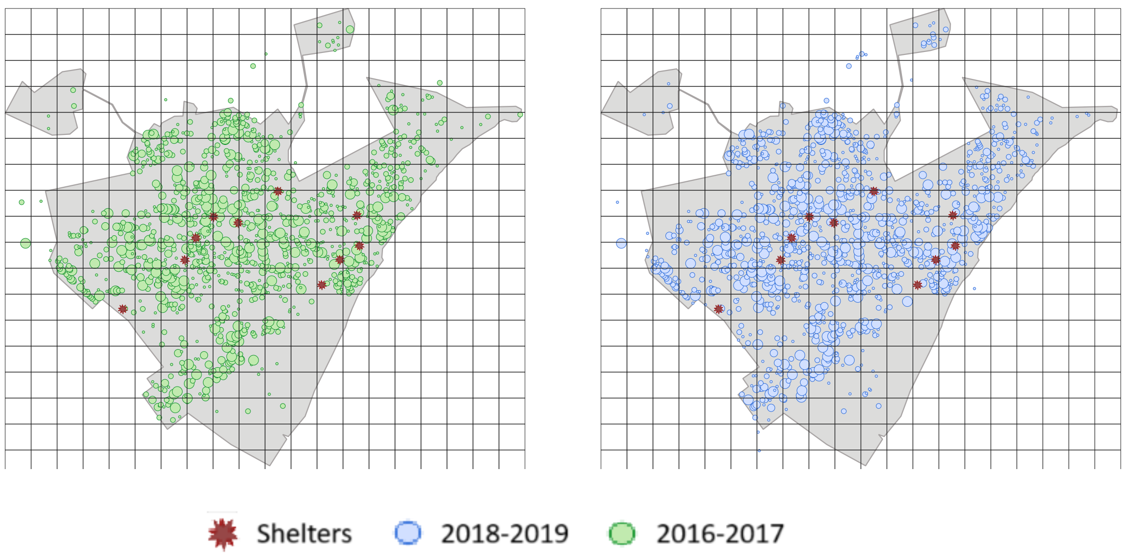
Crime is commonly mentioned in the media, surveys, and politicians' speeches as one of the main concerns associated with migration inflows. To assess whether shelters affected crime, I aggregate crime report data at the polling station level to align with the geographic structure of the voting data. Specifically, each crime report is assigned to its nearest polling station. I then estimate the following TWFE specification using a Poisson regression:

$$\text{Crime}_{jst} = \beta \; Treated_j \times Post_t + FE + Controls_{jst} + \nu_{jst} \tag{3}$$

Crime$_{jst}$ is the total number of crimes of type s (homicides, robbery, or assault) registered in closer to polling station j in quarter t. $Treated_j$, similarly to the voting analysis, is a dummy variable indicating whether polling station "j" is less than 1 kilometer away from the closest Venezuelan refugee shelter. $Post_t$ is a dummy for after the second quarter of 2018. To capture the differential trend in suburban and more central areas of the city, I added the distance to downtown interacted with dummies of quarter ($Controls_{jst}$). Fixed effects are a combination of time, polling station, type, or polling station-type. I also explore a quarter number (1, 2, 3, or 4) interacted with polling station j dummies to capture potential seasonality patterns.

Figure 8: Crime Map Before and After Shelters

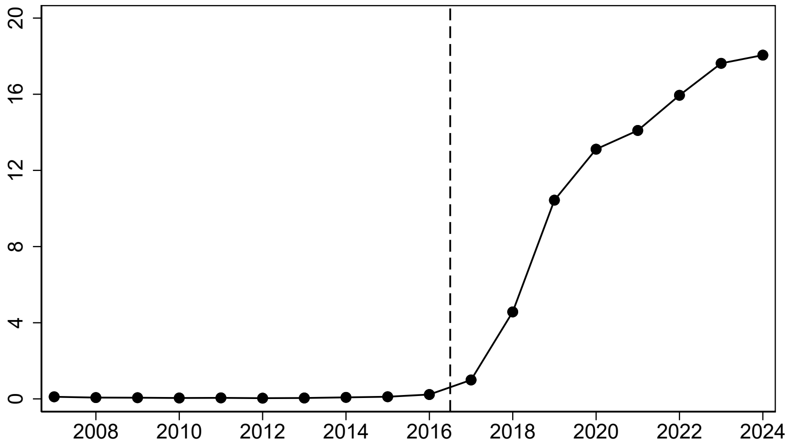
(a) 2016-2017 (b) 2018-2019



Notes: To maintain a consistent two-year time window across both maps, data from 2020 to 2022 are excluded from this plot.

4.3 Empirical Strategy: Public Education

Figure 10: % Venezuelan Students in Boa Vista's Public Schools



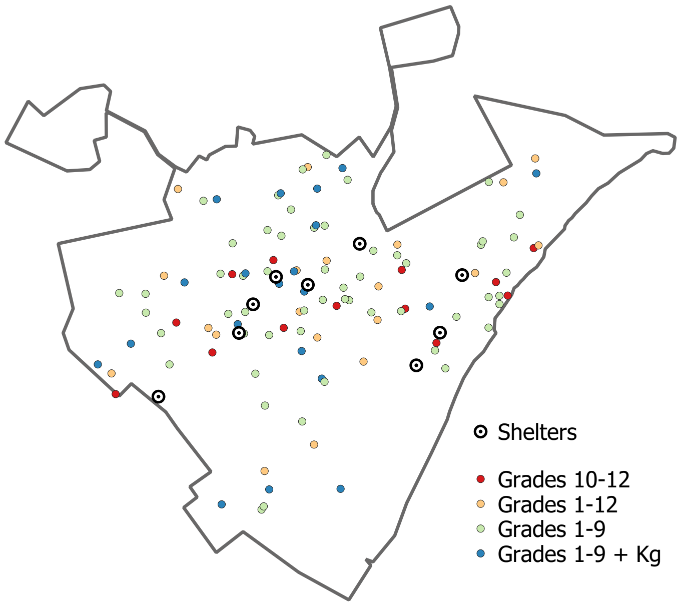
Notes: The sample consists of all urban public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

From 2017 to 2024, the number of Venezuelan students enrolled in public schools in Boa Vista increased more than 20 times, reaching almost 20% of the student population. I use the public school yearly data from 2010 to 2024 to verify the effect of shelters on student composition, congestion, and infrastructure. In particular, I estimate the following event study specification using a panel of schools:

$$Y_{st} = \alpha + \sum_{\tau=2010}^{2015} \beta_{pre} \tau \, Treated_s + \sum_{\tau=2017}^{2024} \beta_{post} \tau \, Treated_s + \gamma_s + \gamma_t + \varepsilon_{st} \tag{4}$$

Y_{st} is the outcome for schools s (such as average classroom size, students-per-teacher ratio, and share of Venezuelan students) in year t. $Treated_s$ is a dummy for whether the school is less than 1 kilometer away from a shelter, respectively. γ_s and γ_t are the school and year fixed effects. Given that most polling stations are public schools, the school-level data is very similar to an aggregation of schools at the polling station level. See the schools and shelters' location in Figure 11.

Figure 11: Schools and Refugee Shelters - Boa Vista (2019)



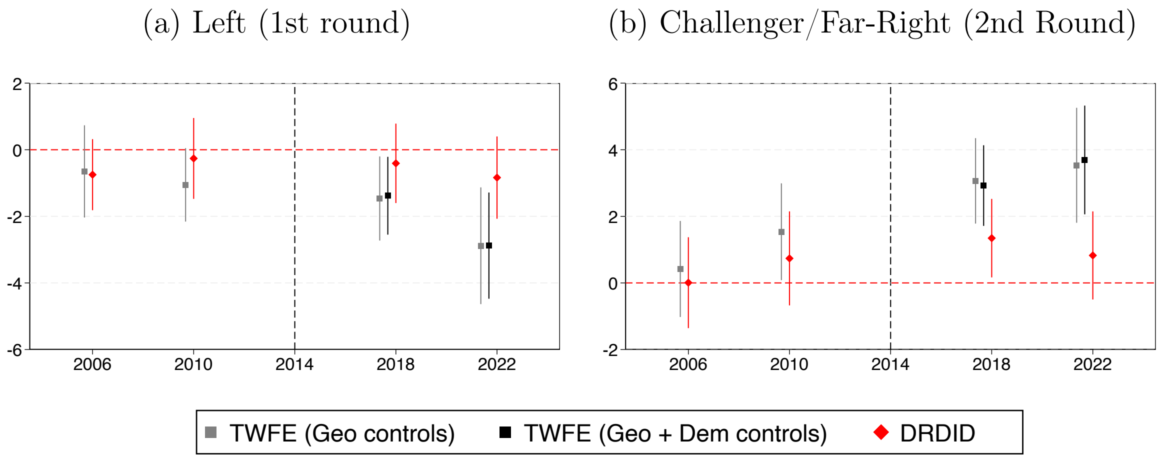
5 Results

5.1 Results: Voting

Presidential Election

According to Figure 12, the left candidate (Haddad) was negatively affected (by 2 to 4 percentage points) in the 2018 second round. Since only two candidates were in the second round, the negative effect on Haddad translates into a positive effect for the Far-Right candidate (Jair Bolsonaro). Finally, the results persist for the 2022 election.

Figure 12: Presidential Election Results - Event Study



Notes: Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. *Challenger* = voting for the party not *Left* (PT).

Table 8: Presidential Election Results

		Y = Left (1st Round)		
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)
Treated × Post	-1.605***	-2.074***	-0.622	-0.563
	(0.539)	(0.630)	(0.685)	(0.558)
	[0.551]	[0.709]	[0.831]	[0.549]
Observations	1,190	714	1,190	1,190
Mean(Y)	19.42	17.35	19.42	19.42
R-squared	0.849	0.902		0.580
		Y = Challenger (2nd Round)		
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)
Treated × Post	2.647***	3.280***	1.086	1.387**
	(0.567)	(0.634)	(0.663)	(0.650)
	[0.708]	[0.754]	[0.879]	[0.851]
Observations	1,190	714	1,190	1,190
Mean(Y)	72.19	75.35	72.19	72.19
R-squared	0.858	0.901		0.679
Geographic Controls	Y	Y	Y	Y
Demographic Controls	N	Y	Y	Y

Notes: Robust standard errors, clustered at the polling station level and neighborhood, are reported in parentheses and brackets, respectively. *** p<0.01, ** p<0.05, * p<0.1.

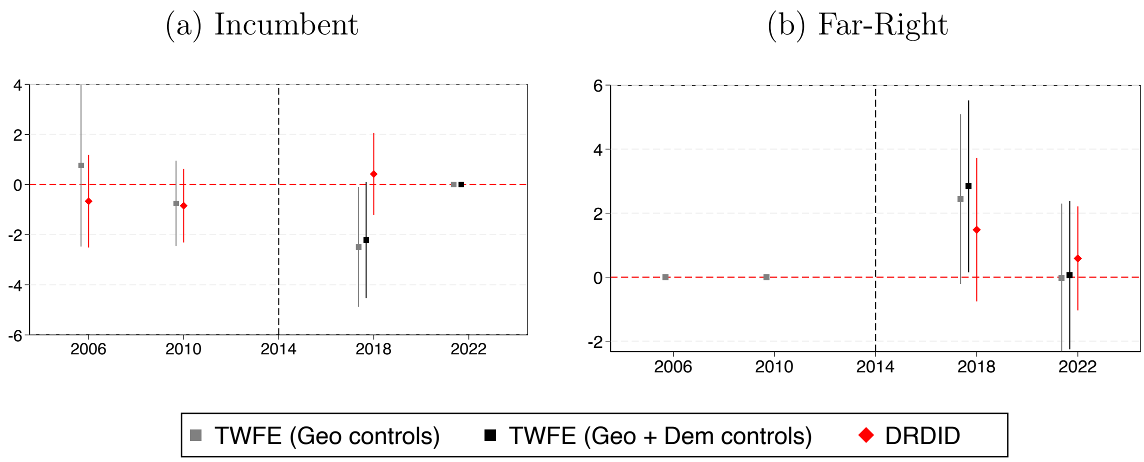
Table 9: President Election Results - Continuous Treatment

	Y = Left (1st Round)			Y = Challenger (2nd Round)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	-2.349***	-3.931**		3.580***	6.195***	
	(0.688)	(1.752)		(0.736)	(1.913)	
Treated(1-2Km) × Post		-2.623			3.843*	
		(1.832)			(1.983)	
Treated(2-3Km) × Post		-0.389			1.917	
					(2.005)	
Z(Distance) × Post			1.330***			-1.901***
			(0.423)			(0.462)
Observations	558	714	714	558	714	714
R-squared	0.905	0.906	0.906	0.908	0.906	0.905
Mean of Outcome	17.46	17.35	17.35	75.34	75.35	75.35
Geographic Controls	Y	Y	Y	Y	Y	Y
Demographic Controls	Y	Y	Y	Y	Y	Y

Notes: Column (1) estimation excluding control pooling stations with less than the median distance to the closest shelter. Robust standard errors, clustered at the polling station, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 14 summarizes the main results of the Governor's election. According to the estimates, there is suggestive evidence that the incumbent governor (Suely) lost between 2 and 4 percentage points of the valid votes in sections within treated polling stations. This incumbent "punishment" accountability result is interesting, given that even though Suely participated in the "Operação Acolhida" effort, she engaged in anti-migration proposals during the 2018 campaign (tried to close the state's border and restrict refugees' access to public services).

Figure 14: Governor Election - Event Study



Notes: Far-Right Party (PSL) did not participate or support any candidate in the 2006 and 2010 state elections. Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD.

Table 10: Governor Election Results

	Y = Incumbent			
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)
Treated × Post	-2.493***	-2.214*	0.422	-0.434
	(0.925)	(1.156)	(1.077)	(2.017)
	[1.055]	[0.866]	(0.677)	(1.769)
Observations	952	476	952	918
Mean(Y)	40.27	26.47	40.27	40.27
R-squared	0.963	0.975		0.931
	Y = Far-Right			
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)
Treated × Post	1.212	1.554	1.033	1.228
	(1.138)	(1.150)	(1.316)	(1.495)
	1.180	1.190	(1.355)	(1.512)
Observations	714	714	714	626
Mean(Y)	44.50	44.50	44.50	44.50
R-squared	0.787	0.796		0.732
Geographic Controls	Y	Y	Y	Y
Demographic Controls	N	Y	Y	Y

Notes: Robust standard errors, clustered at the polling station level and neighborhood, are reported in parentheses and brackets, respectively. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Governor Election Results - Continuous Treatment

	Y = Incumbent			Y = Far-Right		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	-1.888	-5.572***		1.697	4.129**	
	(1.243)	(1.983)		(1.273)	(1.764)	
Treated(1-2Km) × Post		-4.157*			3.137*	
		(2.107)			(1.640)	
Treated(2-3Km) × Post		-2.951			2.912	
		(2.278)			(1.789)	
Z(Distance) × Post			1.408***			-0.939**
			(0.461)			(0.432)
Observations	372	476	476	558	714	714
R-squared	0.977	0.976	0.976	0.797	0.799	0.797
Mean of Outcome	26.32	26.47	26.47	44.61	44.50	44.50
Geographic Controls	Y	Y	Y	Y	Y	Y
Demographic Controls	Y	Y	Y	Y	Y	Y

Notes: Column (1) estimation excluding control pooling stations with less than the median distance to the closest shelter. Robust standard errors, clustered at the polling station, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The incumbent's voting loss translated into increased support for the far-right. There is no longer statistically different from zero in the 2022 election. This result goes in the same direction as other papers in the literature that found positive causal effects of exposure to immigrants on vote shares for right and far-right parties.

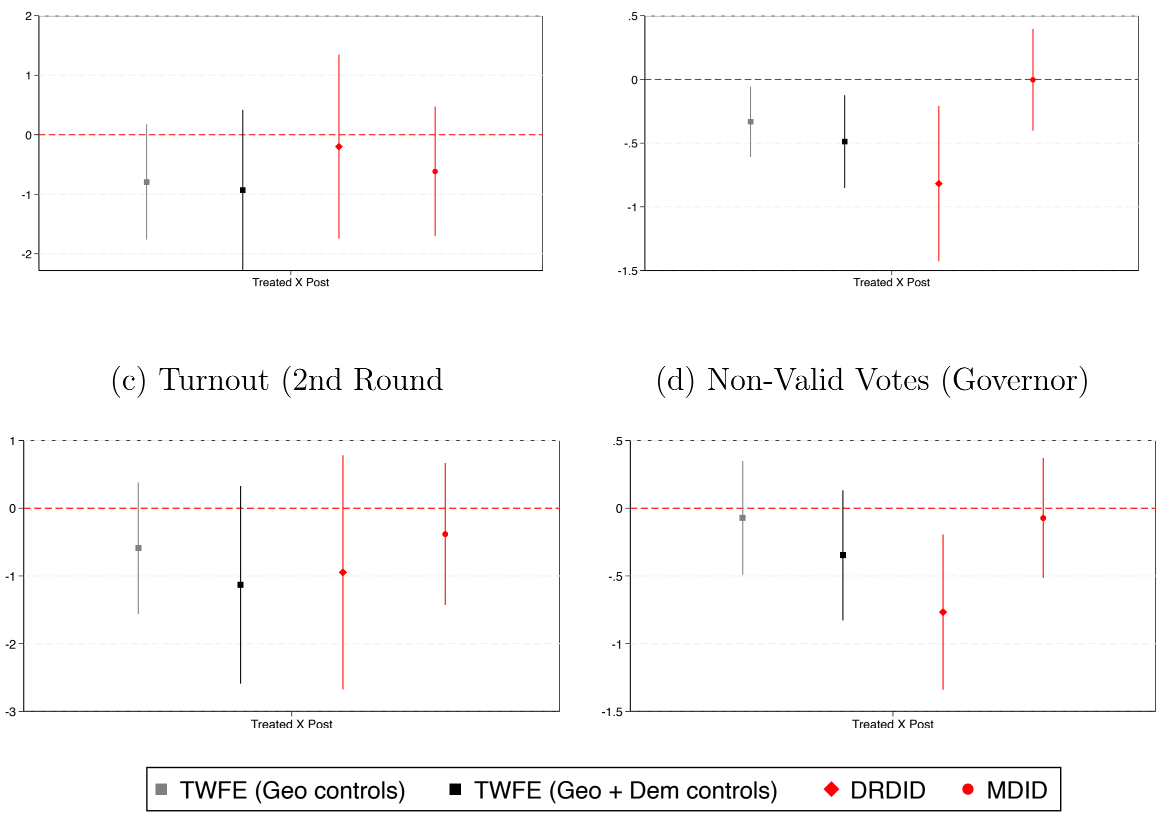
Turnout and Non-Valid Votes

Turnout and non-valid votes could explain the results for the governor and presidential elections. In other words, the shelters could have triggered voters who normally don't show up to vote (turnout increase) or voters who usually don't choose a candidate to select one (decrease share of non-valid votes). According to Figure 16, we don't observe any consistent non-zero effect on the share of non-valid votes. Additionally, the results for turnout rates are noisier, and their statistical significance is inconsistent across the different specifications.

Figure 16: Turnout and Non-Valid Votes Results

(a) Turnout (1st Round)　(b) Non-Valid Votes (President)

(c) Turnout (2nd Round　(d) Non-Valid Votes (Governor)



Notes: Dependent Variable = % of registered voters attending elections or share of non-valid votes among total votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD.

Party Affiliation

Voting for a particular candidate is only one way of expressing political preferences in a democratic system. Citizens can also become members of a party and run for office. To further analyze the political consequences of urban refugee shelters, I explore party affiliation data, which has not yet been explored by the political economy of migration literature. Brazil´s party affiliation is among the highest across democracies at 10% of the adult population (2018) and Boa Vista registered 1,243 new affiliates (all parties) in 2018, an average of 8.75 per polling station.

The data on political affiliation organized by the Superior Electoral Court (TSE) provides individual affiliation information, including the gender, age, party, dates of affiliation, and polling station and room identifiers where the affiliate votes. Following the voting empirical strategy, I explore a yearly (from 2010 to 2022) panel of polling stations and use as dependent variables the share of voters affiliated to the Workers' Party (PT), Bolsonaro's party (PSL), and the incumbent governor's party (PP).

The results (see Figures 52, 53, and 54 in the Appendix H.2) reveal a null effect of shelters on party affiliation. Therefore, the reception policy's political effects were restricted to voting and didn't trigger a more "extreme" political participation behavior. The event study versions reassuringly present no differential trends before 2018.

Putting the results into perspective

To assess the magnitude of the shelter's effect on elections, I explore a municipality-level aggregated data. The goal is to investigate whether the shelters' impact was responsible for leading Boa Vista to a shift to the far-right and benchmark the results with estimates from an aggregated data commonly used by most papers in the literature. I estimate a synthetic control and synthetic Diff-in-Diff exploring a national municipality panel and using other Brazilian municipalities outside Roraima as the control donor pool.[30] Given the computational cost of leading to a large number of potential controls (more than 5,500 municipalities), I use three different smaller samples: a random sample of 300 municipalities, 300 municipalities with the closest pre-2018 average population to Boa Vista, and 300 municipalities with the closest pre-2018 average GDP per capita to Boa Vista.

According to the results (see Figure 18 below and Figures 56, 57, and 58 in the Appendix I), Boa Vista's increase in far-right vote and the decrease in support for the workers' party was also observed in other municipalities of the country. The far-right captured voters even in municipalities that were not directly affected by the Venezuelan flow and its reception policy. Consequently, the shelters shaped within city voting, but they did not disproportionately push Boa Vista towards those candidates. Therefore, the within-city shelter variation explored by this paper captures the meaningful exposure to refugees and also shows how aggregate data can hide important nuances in natives' attitudes towards migrants.

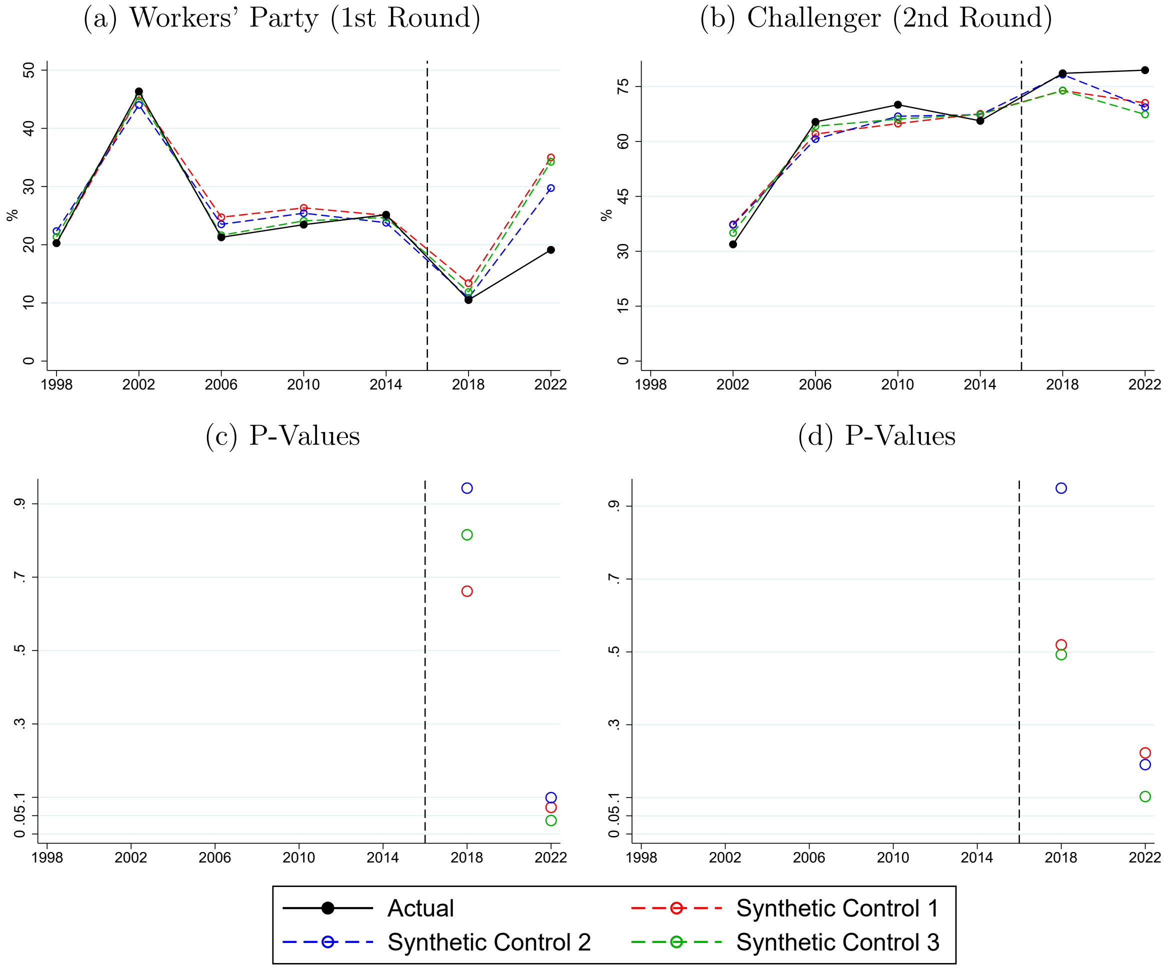
[30]Conventional DiD specifications were also estimated (not reported). However, with a single treated unit (Boa Vista) and the presence of city-specific shocks, event study results indicate that the parallel trends assumption, as expected, is not satisfied.

Table 12: Presidential Election Synthetic Control and Synthetic DiD Results

Dependent Variable:	Y= Left	Y= Challenger
Synthetic Control:		
ATT	-8.910	5.1880
Treatment Effect (2018)	-4.525	0.020
[p-value]	[0.5720]	[0.996]
Treatment Effect (2022)	-13.296^{*}	10.356
[p-value]	[0.0840]	[0.219]
Synthetic DiD:		
ATT	-8.000	5.186
(s.e)	(6.091)	(8.061)

Notes: Synthetic Control and Synthetic DiD p-values and standard errors obtained from iteratively reassigning the treatment to control units where no intervention actually occurred.

Figure 18: Synthetic Control Graphs - Alternative Control Donor Pools



Notes: Synthetic Control 1: random 5% sample of Brazilian municipalities outside Roraima. *Synthetic Control 2*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-2018 average population to Boa Vista. *Synthetic Control 3*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-2018 average GDP per capita to Boa Vista.

5.2 Results: Crime

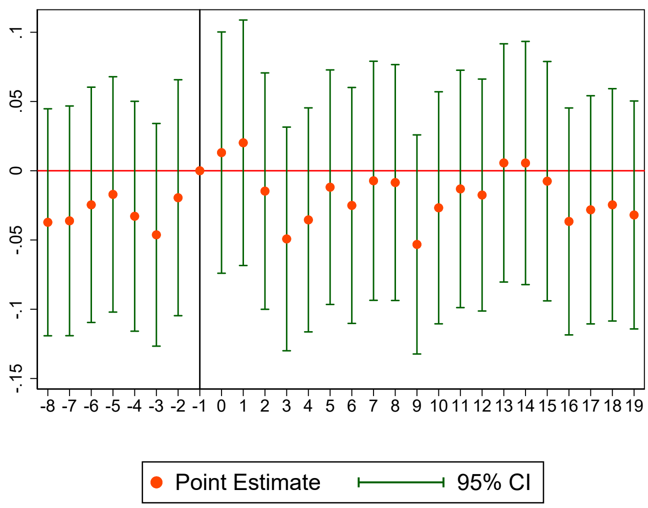
According to the results summarized in Table 13, shelters had no economically relevant and statistically significant effect on criminal activity. The estimates are robust to specifications using different sets of fixed effects and a different data aggregation dividing the city in 1KmX1Km grids. Moreover, we can reject changes of more than 5% of the mean. The event study estimates also reassures parallel trends - see Figure 19.

Table 13: Shelters' Effect on Number of Reported Crimes ($Crime_{jst}$)

	Polling Station Level						Grid-Level
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated × Post	0.040	0.040	0.036	0.036	0.036		-0.038
	(0.061)	(0.061)	(0.061)	(0.061)	(0.061)		(0.079)
Treated(1-2Km) × Post					-0.064		
					(0.059)		
Treated(2-3Km) × Post					0.025		
					(0.080)		
Z(Distance) × Post						0.025	
						(0.036)	
Observations	87,612	81,732	85,260	80,703	80,703	80,703	76,342
Pseudo-R-Squared	0.218	0.208	0.215	0.208	0.208	0.208	0.224
Mean($Crime_{jst}$)	0.275	0.294	0.282	0.298	0.298	0.298	0.315
Time FE	Y	Y	Y	Y	Y	Y	Y
j FE	Y	N	N	N	N	N	N
Crime-Type FE	Y	N	Y	N	N	N	N
j-Crime-Type FE	N	Y	N	Y	Y	Y	Y
Dist. Downtown-Time FE	Y	Y	Y	Y	Y	Y	Y
Quarter-j FE	N	N	Y	Y	Y	Y	Y

Notes: Marginal effects on the number of crimes in brackets using $100 \times (e^{\beta} - 1)\%$. All models include a constant term. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Figure 19: Shelters' Effect on Number of Reported Crimes ($Crime_{jst}$) - Event Study

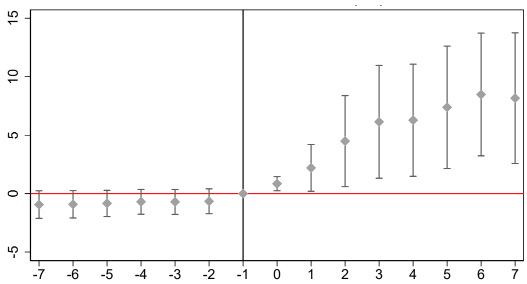


Notes: Event Study based on a "treatment" dummy equal to 1 for a grid with a centroid less than 1 km away from a shelter. The first quarter of 2018 is the baseline period. Estimates from the specification with time and grid-crime-type fixed effects and distance to downtown interacted with time dummies as controls.

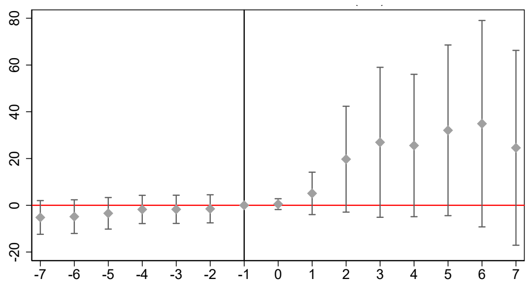
5.3 Results: Public Education

According to the event-study results, proximity to shelters increased the share and number of Venezuelans enrolled - see Figure 20. However, this compositional effect is not accompanied by a congestion effect; the number of students per classroom or the students-to-teacher ratio were not affected. These results indicate that Boa Vista's public education system increased the number of teachers and classrooms in response to the surge in demand, which I confirm with municipality-level data - see Appendix G for details. Given that classroom creation can be achieved at the expense of other infrastructure, I also verified that shelters did not affect the number infrastructure amenities such as reading rooms, libraries, and labs. Finally, shelters had a limited effect on Brazilian students leaving and changing schools, another evidence that discontent is not arising from the usage of this shared public service with Venezuelans.

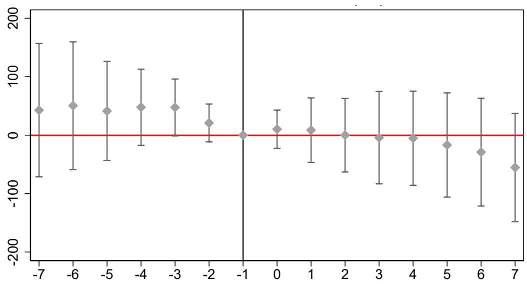
Figure 20: Shelters' Effect on Public Schools



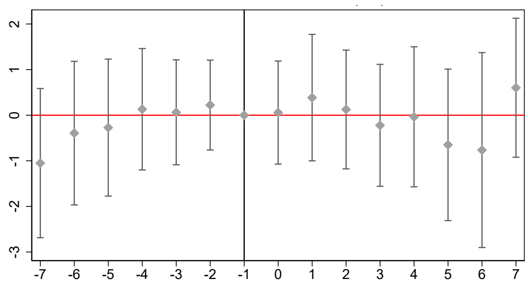
(a) % Venezuelan Students



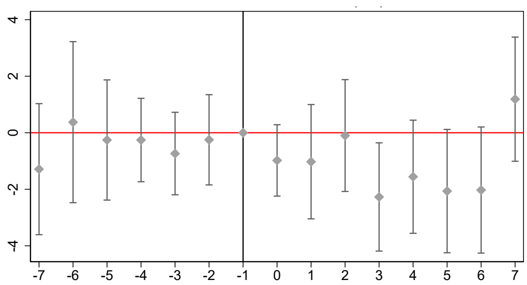
(b) Number of Venezuelan Students



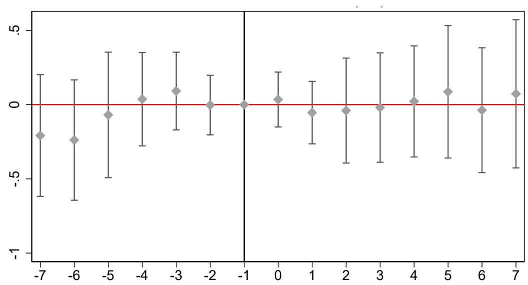
(c) Number of Brazilian Students



(d) Classroom Size



(c) Students per Teacher Ratio

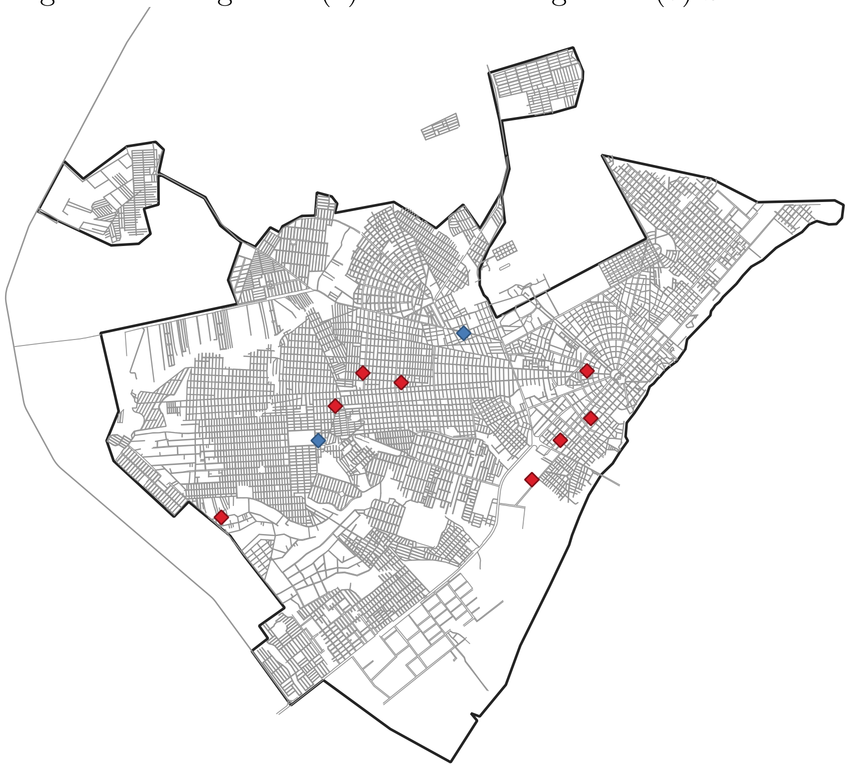


(d) Number Amenities Infrastructure

Notes: Number Amenities Infrastructure = total number of labs, reading rooms, and libraries. Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from a shelter. 2016 is the baseline year. The sample consists of all urban public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

6 Indigenous Vs Non-Indigenous Shelters

Figure 21: Indigenous (2) and Non-Indigenous (7) Shelters Map



The entrance of Venezuelan Indigenous groups was also registered at Roraima's border. They are from multiple ethnicities with no prior history in Brazil. The Warao, or "people of the water," from the Orinoco River delta in Venezuela, are the main group (more than 60%).[31] They relied on fishing, agriculture, and crafts, and mentioned the political and economic crisis and environmental and climate-related reasons (flooding, water contamination, and heavy rains) for leaving their territories in Venezuela. [32]

Indigenous and non-indigenous Venezuelans were housed in separate shelters due to distinct cultural and population profiles and for logistical reasons - see UNHCR Fact Sheet report (2024). Among the 11 shelters, two are designated exclusively to host Venezuelan Indigenous refugees - see Figure 21. Table 14 describes the main demographic and socio-economic differences between indigenous and non-indigenous shelters using the UNHCR shelter reports for October 2018. Indigenous migrants were

[31]Other groups include the Taurepang, Pemón, Arekuna, and many more, with over 13 Indigenous ethnicities registered across Brazil - see the "Warao Refugees in Brazil Report" by UNHCR

[32]For more, see IOM Report.

younger on average, with a larger share of school-age individuals. Moreover, they were less educated, exhibiting an adult illiteracy rate more than five times higher than that of the non-Indigenous sheltered population. The two groups of refugees also present important differences in integration measures. Sheltered Indigenous Venezuelans exhibit lower vaccination rates and possession of documents. The UNHCR also produced a report describing reported vulnerability points (such as homelessness concentration and cases of child labor) in Boa Vista in June 2018. Indigenous shelters are closer on average to those identified points than non-indigenous ones. 45% of all identified vulnerability points (a total of 26) are within 1 km of a shelter, and 50% of those points are closer than 1 km from an indigenous shelter (even though there are only two indigenous shelters and nine non-indigenous).

Table 14: Sheltered Venezuelan Refugee Population Variables (October 2018):

	Indigenous Shelters	**Non-Indigenous Shelters**
Hosted Population*	1,236	2,636
Capacity	109%	87%
Share Male	51,5%	52,4%
Share Some College	9,0%	12,7%
Share High School	46,2%	66,3%
Share Less than High-School	28,9%	18,3%
Share Illiterate	15,9%	2,7%
Share Children (0-11 Years Old)	34,2%	29,6%
Share Teenagers (12-17 Years Old)	11,1%	8,9%
Share Male 60+ Years Old	3,0	1,2
Share Male 18-59 Years Old	50,2%	58,2%
Share Female 60+ Years Old	3,2	1,6
Share Female 18-59 Years Old	52,9%	61,4%
Vaccination Rate	24.4%	64%
Has Social Security Number	35.5%	74.8%
Has work permit card (18+)	25%	44%
"Vulnerability Points" $<$ 1 Km per Shelter **	3	0.67

Notes: (*) considering all operating shelters. The remaining variables are available for seven out of the eleven shelters (Jardim Floresta, Nova Canaa, Pintolandia, Rondon 1, Rondon 3, São Vicente, and Tancredo Neves). (**) "Vulnerability Points" are: homeless concentrations, reported areas of child labor, or areas with families living in vulnerable situations such as informal housing or overcrowded spaces. *Source*: UNHCR and Federal Emergency Assistance Committee reports for June and October 2018.

To evaluate whether there is an additional effect for being exposed to this subgroup of the refugee population, I first ran the following specification:

$$Y_{ijt} = \beta_1 \; Treated_j \times Post_t + \beta_2 \; Treated_j \times Ind.\; Shelter_j \times Post_t + \gamma_i + \mu_t + Controls_{ijt} + \nu_{ijt} \tag{5}$$

Treated$_j$ is a dummy variable indicating whether polling station "j" is less than 1 kilometer away from any of the shelters, and *Ind. Shelter*$_j$ is a dummy for being closer than 1 km to an Indigenous Shelter. Therefore, β_2 captures the additional effect of being close to an indigenous shelter compared to just being exposed to any shelter.

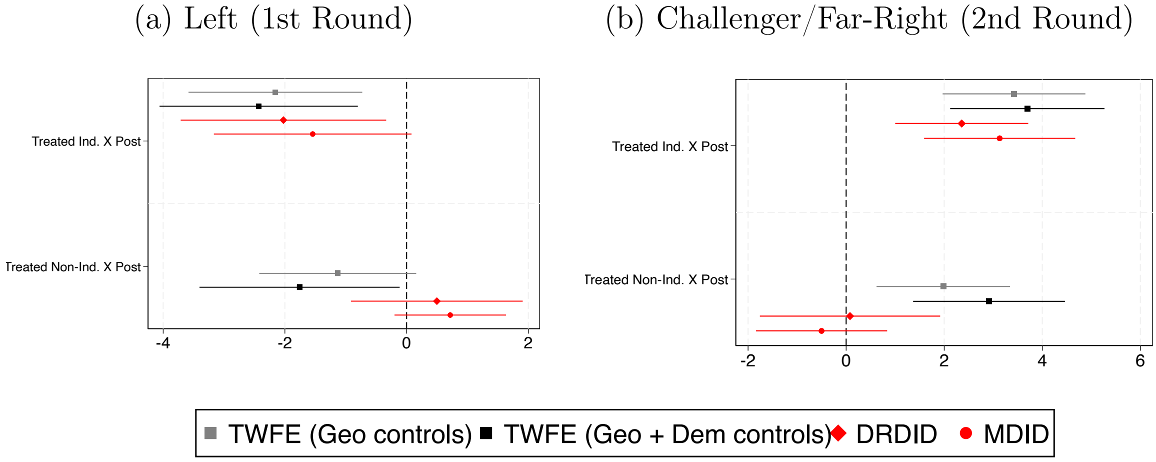
To measure the effects of each type of shelter and account for the fact that some polling stations are treated by both, I estimate the following specification, separating the treatment dummy (Treated$_j$) from equation (1):

$$Y_{ijt} = \beta_1 \; Treat\text{-}Ind._j \times Post_t + \beta_2 \; Treat\text{-}Non\text{-}Ind._j \times Post_t + \gamma_i + \mu_t + Controls_{ijt} + \nu_{ijt} \tag{6}$$

Treated-Ind$_j$ and *Treated-Non-Ind*$_j$ are dummy variables indicating whether polling station "j" is less than 1 kilometer away from the closest indigenous and non-indigenous Venezuelan refugee shelter, respectively.

The results (see Figures **??**, **??**, and **??**) confirm that indigenous shelters are the ones driving the results for both the governor and presidential elections. Therefore, the establishing an urban refugee shelter itself might be less important for political backlash than the cultural, socioeconomic, and integration features of the housed population.

Figure 22: President Election Results by Shelter Type



Notes: Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD. *Challenger* = voting for the party not *Left* (PT).

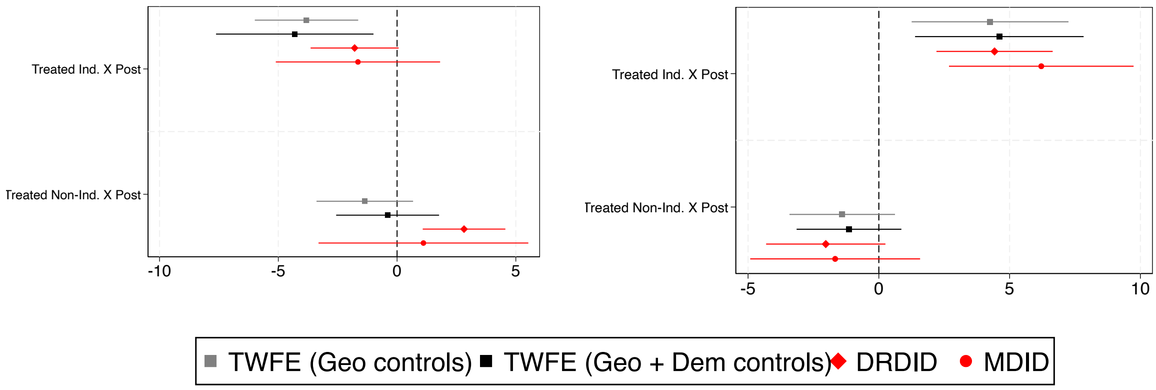
Table 15: Presidential Election Results by Shelter Type

	Y = Left (1st Round)					
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)	(DRDiD)	(MDiD)
Treated × Post	-1.759**					
	(0.821)					
Treated Indigenous × Post	-0.672	-2.430***	-2.024**	-1.543*		
	(1.058)	(0.814)	(0.861)	(0.807)		
Treated Non-Indigenous × Post		-1.759**			0.497	0.717
		(0.821)			(0.720)	(0.456)
Observations	714	714	990	990	985	985
	Y = Challenger (2nd Round)					
Treated × Post	2.913***					
	(0.773)					
Treated Indigenous × Post	0.781	3.694***	2.357***	3.130***		
	(0.916)	(0.786)	(0.692)	(0.766)		
Treated Non-Indigenous × Post		2.913***			0.079	-0.501
		(0.773)			(0.938)	(0.666)
Observations	714	714	990	990	985	985

Notes: Robust standard errors, clustered at the polling station, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 24: Governor Election Results by Shelter Type

(a) Incumbent (b) Far-Right



Notes: Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

Table 16: Governor Election Results by Shelter Type

	Y = Incumbent					
	(TWFE)	(TWFE)	(DRDiD)	(MDiD)	(DRDiD)	(MDiD)
Treated × Post	-0.398					
	(1.082)					
Treated Indigenous × Post	-3.909**	-4.307**	-1.787*	-1.647		
	(1.620)	(1.655)	(0.945)	(1.718)		
Treated Non-Indigenous × Post		-0.398			2.820***	1.110
		(1.082)			(0.888)	(2.199)
Observations	476	476	792	754	788	745
	Y = Far-Right					
Treated × Post	-1.141					
	(1.000)					
Treated Indigenous × Post	5.748***	4.607***	4.425***	6.211***		
	(1.646)	(1.609)	(1.134)	(1.753)		
Treated Non-Indigenous × Post		-1.141			-2.029*	-1.671
		(1.000)			(1.164)	(1.617)
Observations	714	714	594	526	591	523

Notes: Robust standard errors, clustered at the polling station, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

For robustness I explore two alternative data aggregations. First, I aggregate voting outcomes and covariates at the polling station level and adjust the specifications for a panel of polling stations to estimate the results. Second, leveraging the features behind voter allocation, I construct a fake voting district using Voronoi Polygons that partition space into regions based on the nearest polling station. [33] The estimates from both polling stations and polygon panels confirm the section-level results for both the governor and presidential elections (results not reported in this draft).

Given that there are only two Indigenous shelters compared with nine non-indigenous, I also run a randomized inference exercise, drawing different combinations randomly selected 2 shelters to be designated as "indigenous" to obtain $\textit{Treated-Non-Ind}_j$ and $\textit{Dist. Non-Ind.}_j$. According to the results, across different specifications, the estimates using the actual shelter type definition are in the 95% tail of the distribution - see Figures 59, 60, and 61 in Appendix J.

Finally, to rule out that the Indigenous shelters' heterogeneous effect is actually just capturing multiple treatments, I exclude polling stations treated by both types of shelters. The results are noisier but still confirm that locals' political reaction is exclusively caused by proximity to indigenous shelters and not potential proximity to non-indigenous ones - See Table 22 in Appendix J.

Party Affiliation

The results (see Figures 52, 53, and 54 in the Appendix H.2) confirm the null effect of both non-indigenous and indigenous shelters on party affiliation. Therefore, the reception policy's political effects were restricted to voting and didn't trigger a more "extreme" political participation behavior. The event study versions reassuringly present no differential trends before 2018.

Crime

According to the results (see Table 17), neither shelter type affected criminal activity. The estimates are robust to specifications using different sets of fixed effects and are also non-significant when estimating a Poisson regression. The event study version of Equation (1) estimates (see Figure 26) also reassures parallel trends.

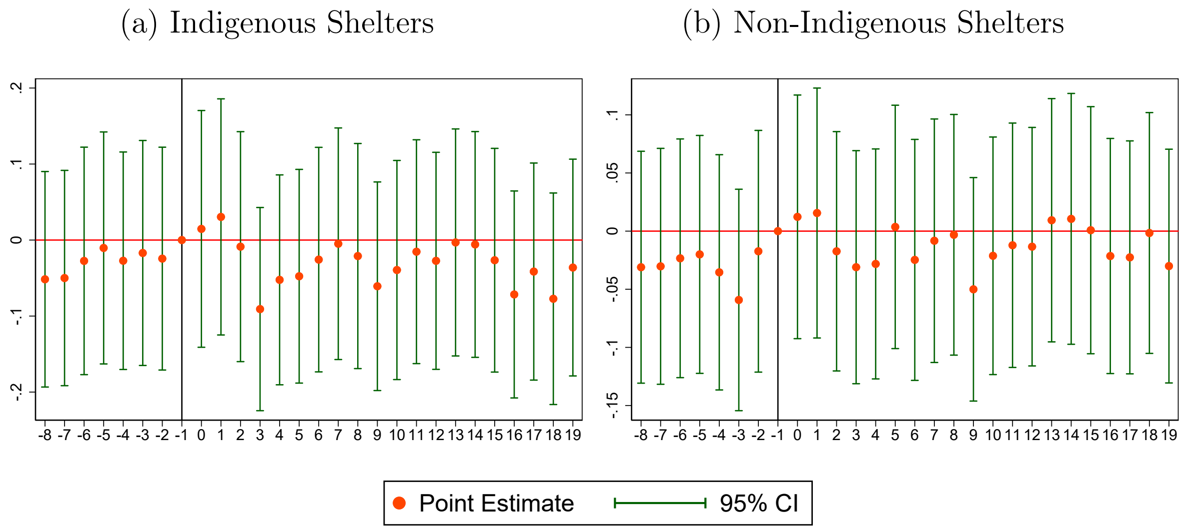
[33] See Appendix F for details.

Table 17: Shelters' Effect on Number of Reported Crimes ($Crime_{jst}$) - Shelter Type

	(1)	(2)	(3)	(4)	(5)	(6)
Treated (Ind. Shelter)*Post	0.025	0.025	0.020	0.020		-0.140
	(0.116)	(0.116)	(0.116)	(0.116)		(0.134)
Treated (Non-Ind. Shelter)*Post	0.045	0.045	0.043	0.043		-0.001
	(0.071)	(0.071)	(0.072)	(0.072)		(0.097)
Z(Distance Ind.)*Post					0.092**	
					(0.042)	
Z(Distance Non-Ind.)*Post					-0.020	
					(0.040)	
Observations	87,612	81,732	85,260	80,703	80,703	76,342
Pseudo-R-Squared	0.218	0.208	0.215	0.208	0.208	0.224
Mean ($Crime_{jst}$)	0.275	0.294	0.282	0.298	0.298	0.315
Time FE	Y	Y	Y	Y	Y	Y
j FE	Y	N	N	N	N	N
Crime-Type FE	Y	N	Y	N	N	N
j-Crime-Type FE	N	Y	N	Y	Y	Y
Dist. Downtown-Time FE	Y	Y	Y	Y	Y	Y
Quarter-j FE	N	N	Y	Y	Y	Y

Notes: Marginal effects on the number of crimes in brackets using $100 \times (e^{\beta} - 1)\%$. All models include a constant term. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Figure 26: Shelters' Effect on Number of Reported Crimes ($Crime_{jst}$) - Event Study



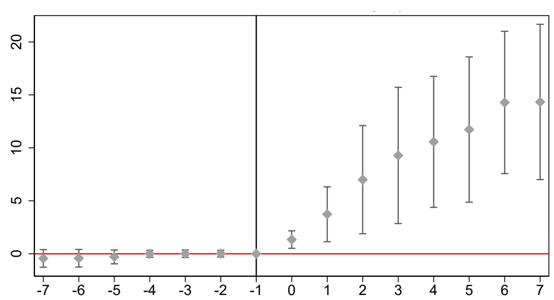
Notes: Event Study based on a "treatment" dummy equal to 1 polling station less than 1 km away from Indigenous or Non-Indigenous shelters. The first quarter of 2018 is the baseline period. Estimates from the specification with time and polling station-crime-type fixed effects and distance to downtown interacted with time dummies as control.

Public Education

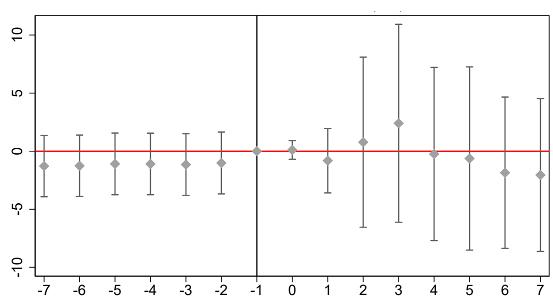
The event study estimates from equation (2) for the share of Venezuelan students and classroom sizes are plotted in Figure 28. Despite hosting a larger share of school-age Venezuelans, Indigenous shelters do not affect the surrounding schools' share or number of Venezuelan students. Contrastingly, non-indigenous shelters increased Venezuelan enrollment in schools less than one kilometer away.[34] Moreover, neither shelter had an effect on different measures of school congestion or infrastructure: number of students per teacher or number of infrastructure amenities such as libraries and laboratories - see Figures 50, and 51 in the Appendix H.

Figure 28: Shelters' Effect on Schools - Shelter Type

(a) % Venezuela Students

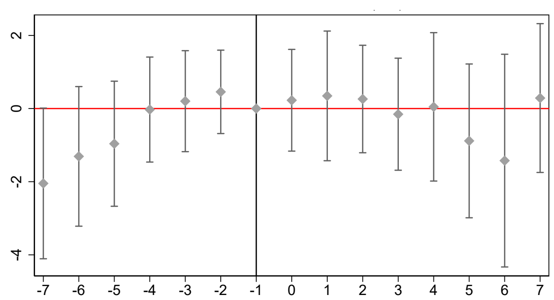


(a.1) Non-Indigenous Shelters

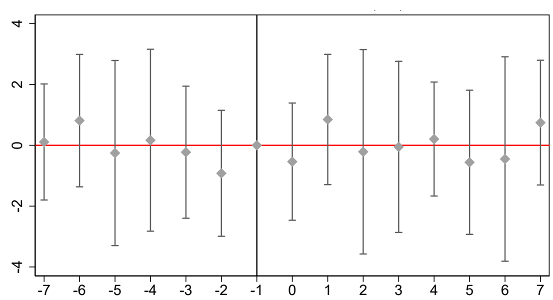


(a.2) Indigenous Shelters

(b) Classroom Size



(b.1) Non-Indigenous Shelters



(b.2) Indigenous Shelters

Notes: Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from an Indigenous shelter. 2016 is the baseline year. The sample consists of all public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

[34]See Figures 48 and 49 in Appendix H, for results on number of Brazilian and Venezuelan students.

7 Conclusion

Since 2014, over one million Venezuelans have entered Brazil, making the country's northern bordering state of Roraima the center of an unprecedented forced migration inflow. Brazil adopted a local integration model by granting refugees extensive rights and access to public services and opening urban shelters in Roraima's capital. This paper studied how this reception policy, better suited to today's urban and long-term displacement realities, shaped crime, public service congestion (public schools), and voting.

I leverage the quasi-random placement of shelters to estimate a Diff-in-Diff comparing areas close to further away from them. First, I don't observe any adverse effect of shelters on public schools' congestion or crime. Looking at election results, I find that natives closer to refugee shelters increased support for the far-right for both the governor and presidential elections, and penalized the incumbent governor who participated in the reception efforts. The results cannot be explained by changes in voters' composition or election logistics. Moreover, the estimates are robust to different specifications and definitions of treatment.

Using a synthetic control, I verified that the city's overall far-right voting mirrored that observed in other municipalities not affected by the migration flow. Therefore, the effects of shelters were not strong enough to produce a citywide increase in populist far-right voting, and the nuances observed would not have been captured with aggregated municipality-level data commonly used in the literature. Moreover, it confirms that shelters' location captures the relevant variation in exposure to migrants.

Importantly, the results are largely driven by shelters hosting Indigenous Venezuelans. According to UNHCR reports and public school enrollment data, Indigenous refugees are an especially vulnerable and low-integrated subgroup of the refugee population. The salience of refugees' presence and locals' exposure to poverty and vulnerability, such as children outside school, child labor, and homelessness, can be behind the effects. Therefore, ethnic-specific contact triggered political backlash even when we don't observe worsening crime and public goods congestion.

These findings suggest that political reactions to refugee reception can be highly localized and that establishing a refugee shelter may be less important for political backlash than the cultural features of the people housed. Finally, this paper's results also document a trade-off in tailored or culturally specific reception strategies: while they can address the particular needs of certain groups, depending on their logistics, they can also increase perceived social distance and trigger stronger attitude reactions among natives.

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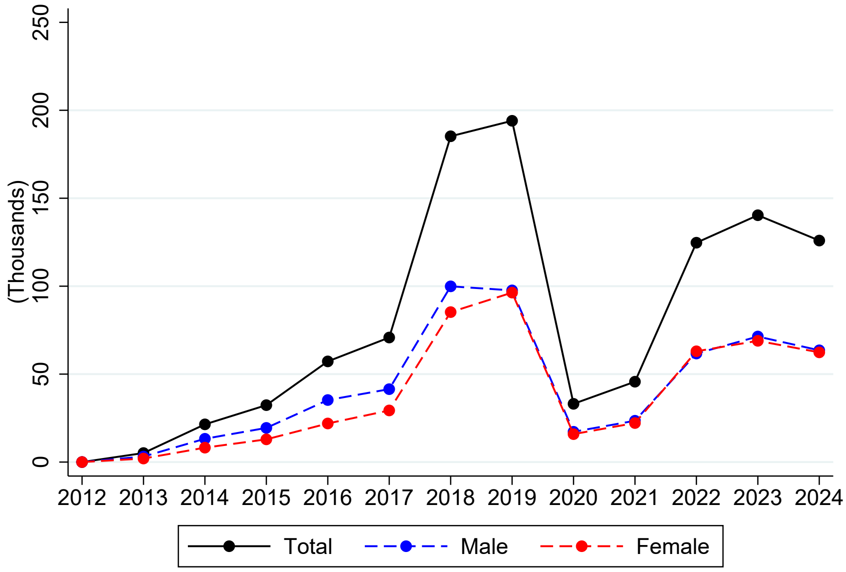
Appendix

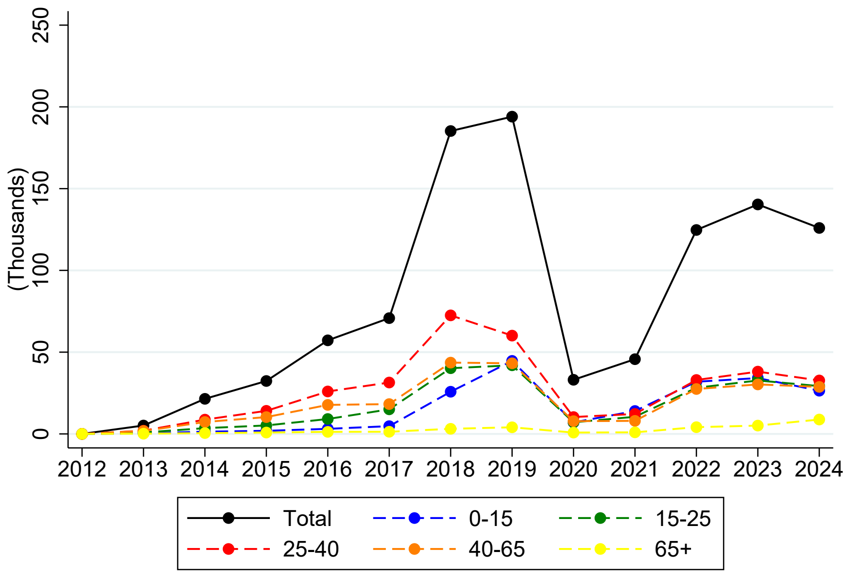
A Venezuelan Refugee Flow

Figure 29: Venezuelan Entrance Flows to Roraima



Source: Sistema de Tráfego Internacional (STI).

Figure 30: Venezuelan Entrance Flows to Roraima



Source: Sistema de Tráfego Internacional (STI).

Figure 31: Sheltered Refugees Vs Roraima's Population - Age

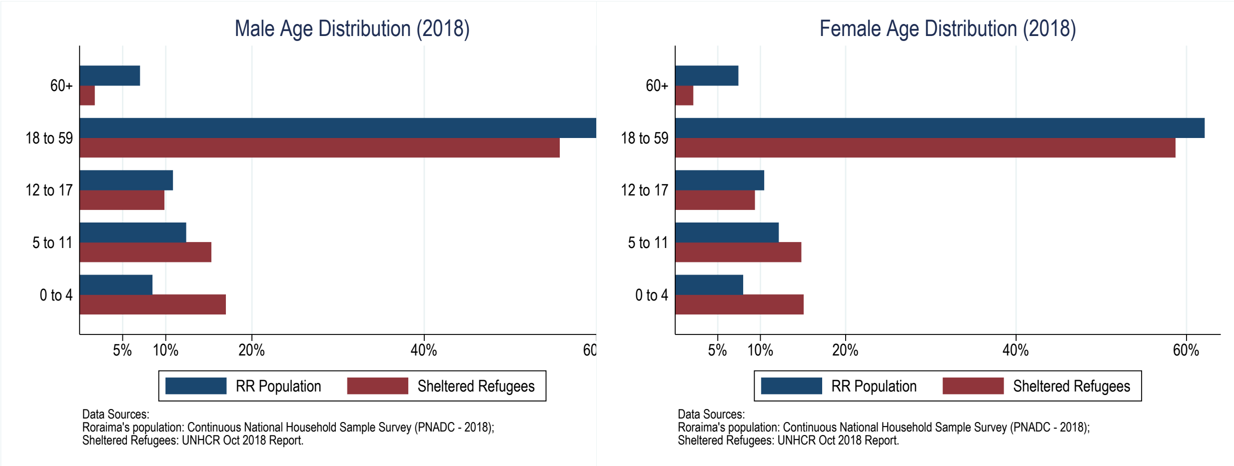


Figure 32: Sheltered Refugees Vs Roraima's Population - Education

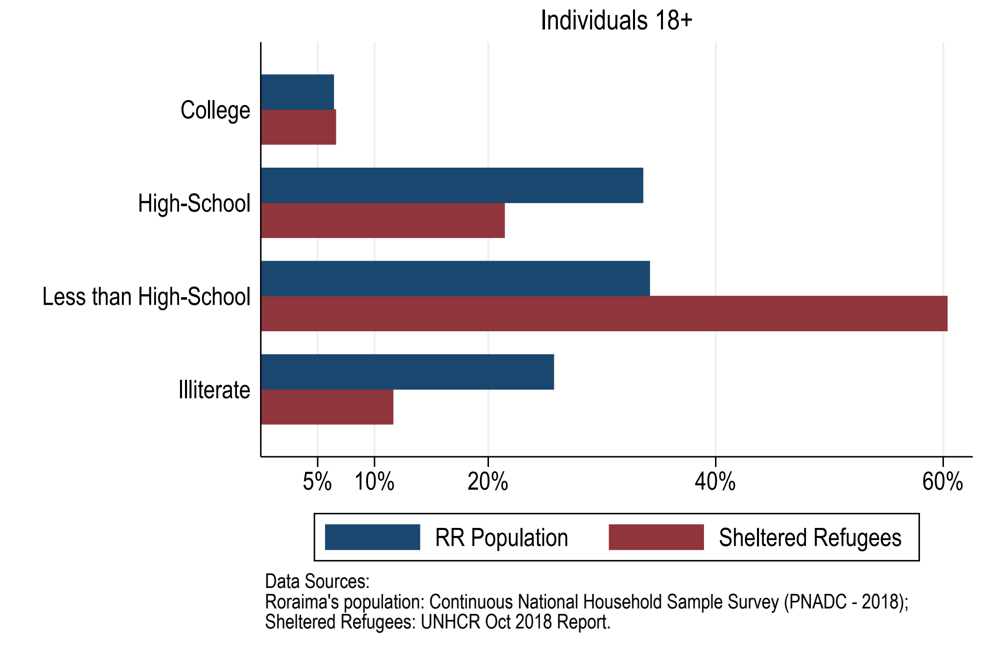
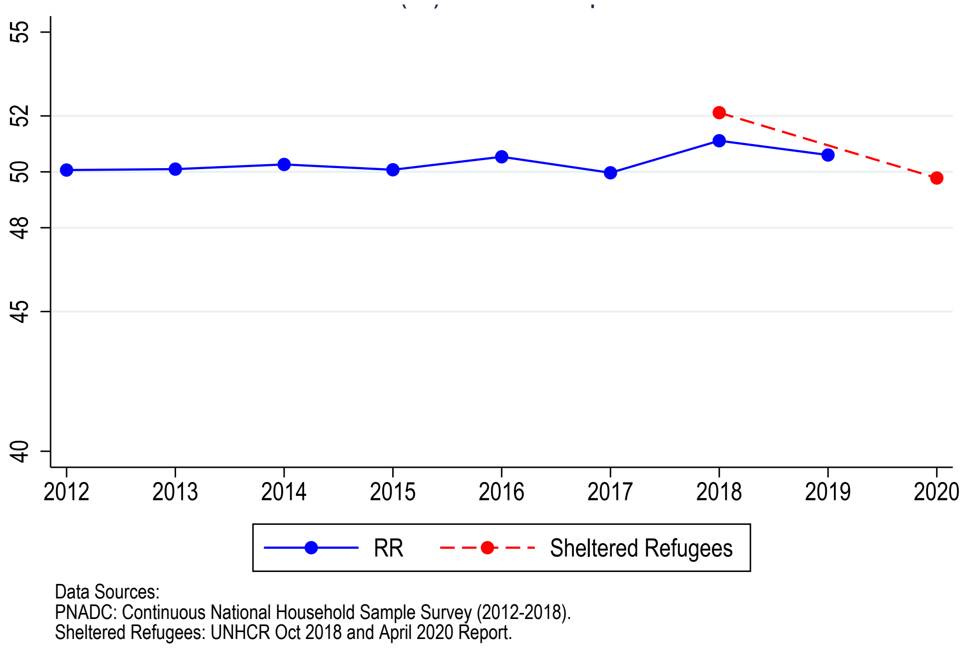


Figure 33: Sheltered Refugees Vs Roraima's Population - Gender (% Male)



B Reception Policy Details

Figure 34: "Operação Acolhida" Anual Budget (Millions R$)

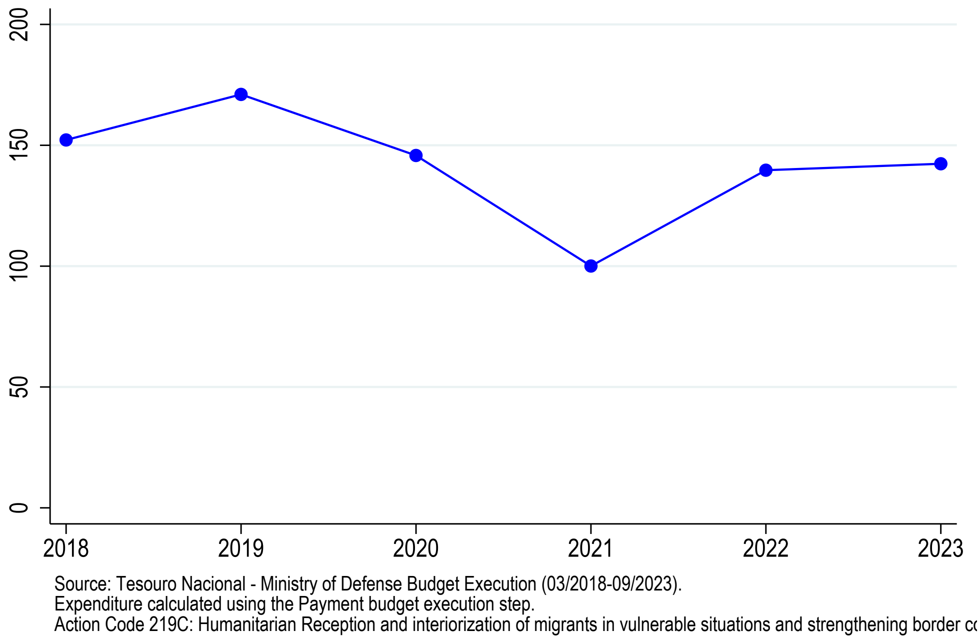


Table 18: Shelters Statistics

Name	Opening Date	Capacity (September or October 2018)	Sheltered Population (September or October 2018)	Capacity (August 2020)	Sheltered Population (September 2020)	Average Length of Stay - days (September 2020)
Pintolândia	March 2018	448	754	640	536	470
Tancredo Neves	March 2018	232	324	280	217	270
Hélio Campos	December 2017	no info	252*	closed	closed	closed
Jardim Floresta	March 2018	594	693	550	368	293
São Vicente	April 2018	378	353	300	251	270
Nova Canaã	April 2018	390	436	350	235	265
Rondon 1	July 2018	600	715	810	559	240
Latife Salomão	April 2018	no info	514*	300	195	248
Santa Tereza	May 2018	no info	531*	320	255	191
Rondon 2	September 2018	no info	453*	645	340	223
Rondon 3	October 2018	1086*	344*	1386	844	245
São Vicente 2	July 2019	did not exist	did not exist	250	110	177

Notes: (*) data not available in UNHCR reports, so obtained from "Operação Acolhida" meeting minutes. The average length of stay is not available for 2018.

C Political Background

Figure 35: Timeline Brazil's Presidents

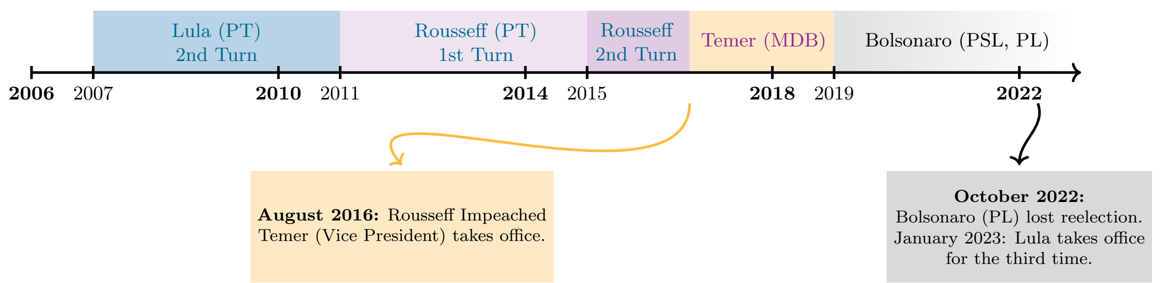
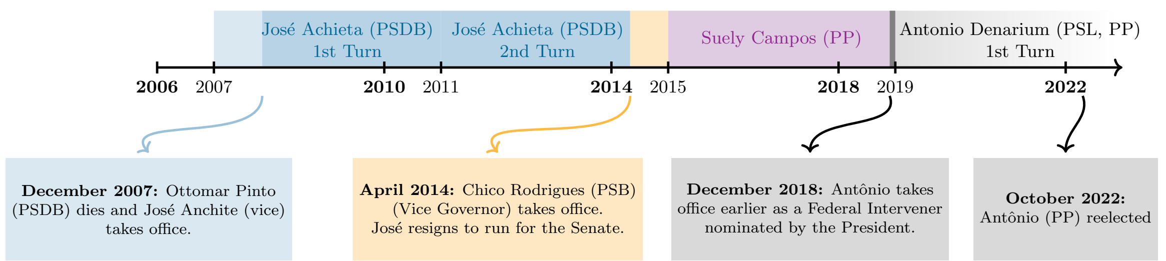


Figure 36: Timeline RR's Government



The relationship between the state and federal government was not only characterized by partnerships and cooperation. Suely (2018 incumbent governor) claimed during the 2018 campaign that the federal government's response to the Venezuelan flow in

Roraima was late and insufficient. Moreover, while Suely wanted to close the border to prevent the entrance of more Venezuelans (she even appealed to the Supreme Court), the President refused to do so, arguing it would violate humanitarian reception principles.[35] Finally, two months before the election, Suely also published an unconstitutional act trying to enhance deportation enforcement and to introduce a passport possession requirement for Venezuelans to access non-emergency public services.[36]

During 2018, Roraima also suffered from a financial crisis and a surge in crime. The prison system was especially vulnerable (overcrowded and under-staffed) and mass escapes and riots were registered in 2018.[37] During the campaign, Suely claimed the former Governor's poor financial management, the unprecedented refugee flow, and the absence of federal government assistance created "the most challenging environment a Roraima's governor ever faced".

Table 19: Governor Election - Parties Classification

	2022	**2018**	**2014**	**2010**	**2006**
Incumbent		PP	PP	PP	PSDB
Far-Right	PP	PSL	PSB	-	-
Anchieta		PSDB	PSB	PSDB	PSDB

Table 20: President Election - Parties Classification

	2022	**2018**	**2014**	**2010**	**2006**
Incumbent	-	-	-	-	-
Far-Right	PL	PSL	PSB	-	PSL
Left	PT	PT	PT	PT	PT

D Latitude and Longitude of Polling Stations

Hidalgo's code output contains a polling station panel ID, the coordinates from different data sources, and also provides a predicted coordinate (useful when coordinates from TSE are not available) based on a model using the TSE data as a benchmark.

[35] *"Governor of Roraima asks to close Brazil's border with Venezuela"*
[36] *"Government of Roraima signs decree that tightens foreigners access to public services"*
[37] *"Roraima's prison system in crisis will be taken over by the federal government"*

It also provides a predicted distance (in Km) between the chosen longitude, latitude, and "true" benchmark longitude and latitude. The following procedures were followed to use and check this data:

1. I kept only observations for Boa Vista (Roraima) municipality.

2. I used the location provided by the TSE available only for 2018 and 2020 for a given panel ID to complete the location information for the previous elections (2006 to 2016). This completed 84.68% of all pooling station-year observations. The remaining 15.32% of the sample are mostly polling stations that didn't exist anymore in 2018 and 2020.

3. I used Hidalgo's predicted location for this 15.32% of the polling station-year sample. It's predicted location searches for the address and name of the polling station in different administrative data, such as the Census and the list of public schools' locations.

4. However, some pooling stations (3.26% of the entire pooling station-year sample) end up presenting different predicted locations depending on the year. This could be because of polling stations' relocation, some error in Hidalgo's panel ID, or different data availability for different years. In those cases, I used the predicted location with the smaller predicted error (therefore, I ignored any potential relocation of polling stations).

5. Then I checked that different polling stations presented different locations. This was the case, as expected, for more than 93% of the sample, however, 6.95% of the sample consisted of different polling stations that shared the same latitude and longitude. This can be explained either by an error in Hidalgo's panel ID or because some geographic coordinate data sources were at a higher geographic level (such as at the census tract level). Therefore, in this case, I searched the address manually using Google Maps and obtained the latitude and longitude.

6. TSE provides two polling station identifiers. However, they do not work as a proper panel ID, given that they can be reused in case a polling station is

destroyed or moved. However, I can use this TSE "quasi-panel ID" to check Hidalgo's panel ID (i.e., no polling stations with different IDs that are the same). This exercise raised an alert for 12.32% of the sample. Among those, 100 observations (8.80% of the sample) were from panel stations that should have the same ID. This occurred mainly because, for some years, addresses were written in different ways (the polling station was at a corner, and each year a different street was used for its address, or the name of the street changed). For this 8.80% of the sample, the coordinate chosen follows the following priority TSE, Google Maps, and Hidalgo Predicted.

See Table 21 below for the final description of the polling stations' geographic coordinates data source.

Table 21: Polling Stations' Geographic Coordinates Data Source

Geo. Coordinate Data Source	**% Sample**	**% Polling Stations**
TSE (Supreme Court for Elections)	87.32%	76.63%
Google Maps	6.60%	10.33%
Hidalgo Predicted	5.28%	11.42%
No Latitude/Longitude	0.79%	1.63%

E Fingerprint scan and Voters' Demographic Info

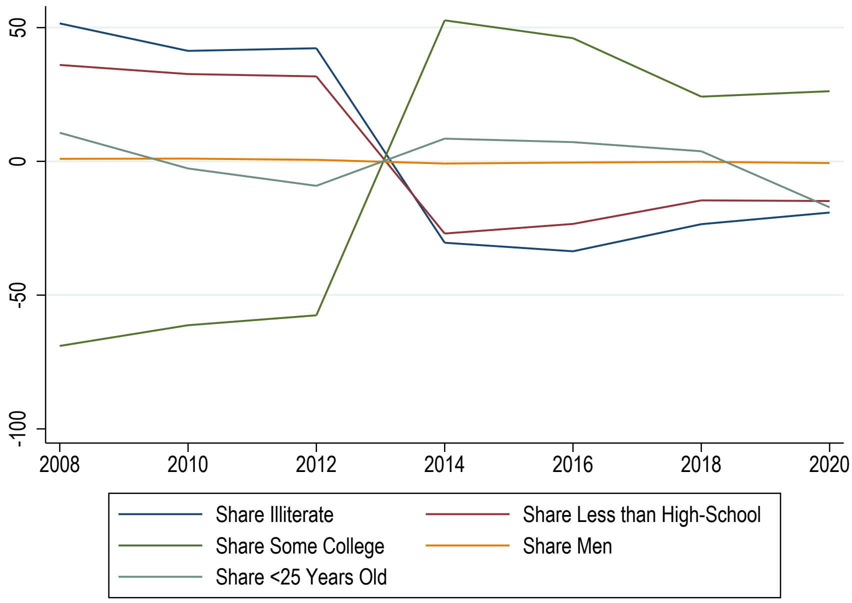
In this appendix section, I investigate how voter fingerprint registration altered the quality of voters' demographic variables. First, I calculated the following yearly index to verify how big the update in voters' demographic variables was after the 2013 fingerprint requirement that made all voters come back to the offices.

$$ID_t = \frac{1}{N}\sum_{i=1}^{N} 100 \times \frac{(y_{ijt} - \bar{y}_{ij})}{\bar{y}_{ij}}$$

$\bar{y}_{ij}$ is the average across elections (2008 to 2020) of outcome y for section "i" ($\frac{\sum_t y_{ijt}}{T}$). Therefore, ID_t represents the average sections' percentage deviation from their 2008-2020 average.

As we can see from Figure 37, ID_t associated with education variables are consistently above zero before 2013 and negative after. Therefore, educational information

Figure 37: ID_t for different variables



seems to have presented important updates after 2013 in the direction of more education. We don't observe this pattern for age or gender info. This could be because gender and age information don't require constant updates from the voters; on the other hand, education can change (upgrade) over time. Given that voters are registering when they are 18 years old, potential late high-school degree acquisition and college attendance were not being captured for a considerable proportion of the voter population.

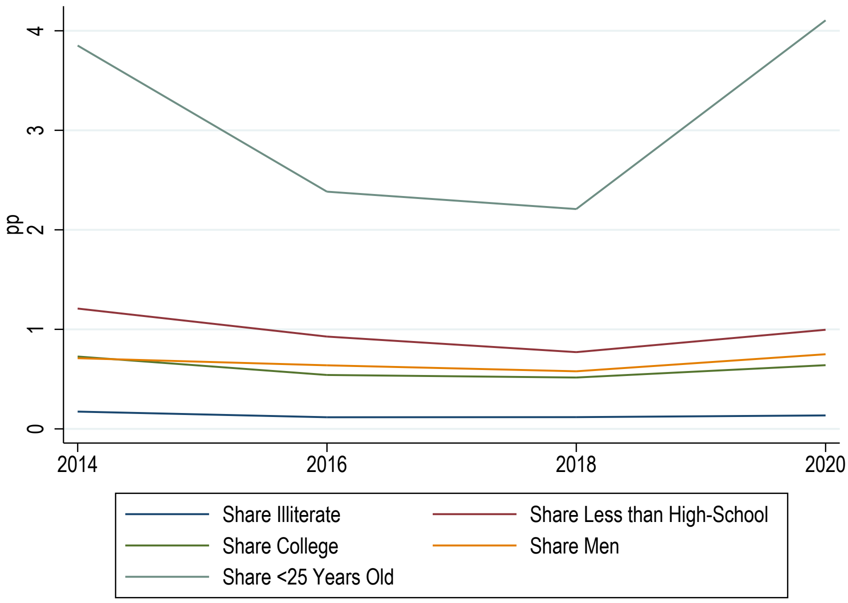
To show that after 2013 the voters' characteristics in each section were stable, i.e. people were not moving between sections or polling stations over elections, and there is an inertia in section assignment (as described by the electoral code), I calculated the following yearly index for $t > 2013$:

$$IA_t = \frac{1}{N}\sum_{i=1}^{N} |y_{ijt} - \bar{y}_{ij}|$$

$\bar{y}_{ij}$ is the average across elections (2014 to 2020) of outcome y for section "i" ($\frac{\sum_t y_{ijt}}{T}$). Given that all outcomes are a share (0 to 100) the IA can be interpreted as a percentage point absolute sections' average deviation.

According to Figure 38, voter demographic information is stable and suffers minor

Figure 38: IA_t for different variables



deviations across elections. This goes in the direction of the National Electoral Code stating that voters will be permanently linked to their original section unless of some specific exceptions.

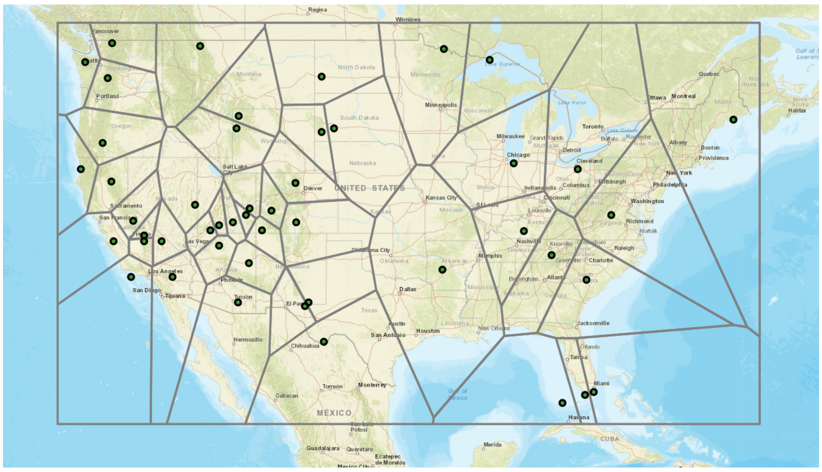
F Voronoi Polygons Panel

Given the desirable electoral code features, designing areas that mimic a voting district is possible. Considering that distance is an important factor during the assignment of polling stations, I will explore Voronoi Polygons (described next) to obtain "fake" voting districts for Boa Vista's urban area for robustness.

Voronoi Polygons are great at dividing the space based on the distance to reference points. The Polygon created around a certain reference point indicates that all individuals living within the Polygon "i" are closer (in terms of distance) to the reference point at the center of "i" than any other reference point. Therefore, more isolated reference points would be associated with a bigger polygon. Figure 39 below shows the Voronoi Polygons constructed using the US National Parks location as reference points. According to the map, someone living in San Francisco is closer in distance to the Pinnacles National Park than any other National Parks (Yosemite and

Yellowstone, for example).

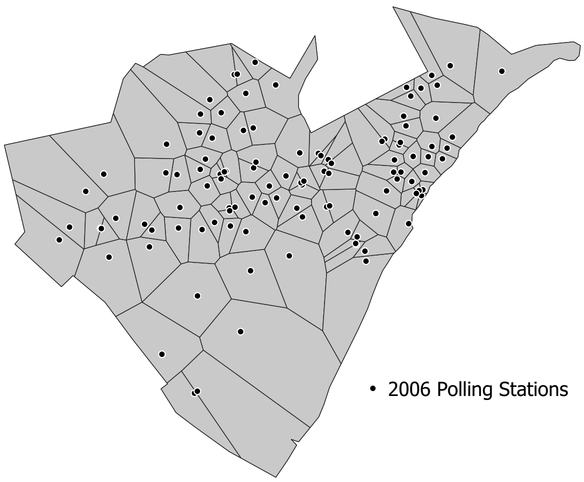
Figure 39: Voronoi Polygons using US National Parks



Note: Image Source.

I used the 2006 polling stations as reference points to obtain the Voronoi Polygons for the entire urban area of Boa Vista (there were no shelters in the municipality's rural part). Boa Vista's urban limit was drawn based on the 2010 Map of streets and avenues by the National Statistics Institute (IBGE). Figure 40 shows all the 111 polygons constructed based on the 2006 polling stations.

Figure 40: Voronoi Polygons using 2006 Polling Stations (Urban Area of Boa Vista)



By construction, the political outcome of observation "i" in 2006 will be measured using the single 2006 polling station data that generated that polygon "i". However, after 2006, there was destruction and the creation of new polling stations. Therefore, a weighting strategy will be necessary, given that more than one polling station might be located within the same polygon after 2006. To get the weights, I will first overlap the Voronoi patterns of 2006 and year "t" for $t > 2006/2008$ (see Figure 41 for the 2006 and 2010 polygons overlap example). The weight that a certain polling station "j" will receive when calculating the outcome in a year "t" for observation/polygon "i" will be equal to the share of j's Voronoi area in the year "t" that lies within observation/polygon "i". The same weighting strategy will be used to obtain "i" covariates (voters' characteristics) over time.

Figure 41: Overlaping the Voronoi Diagrams of 2006 and 2010 Polling Stations

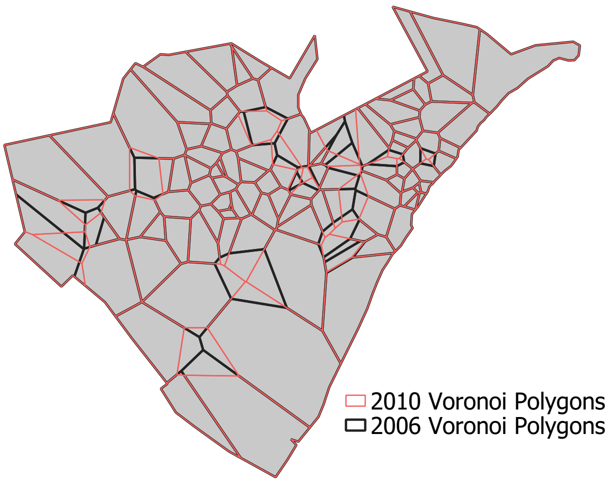
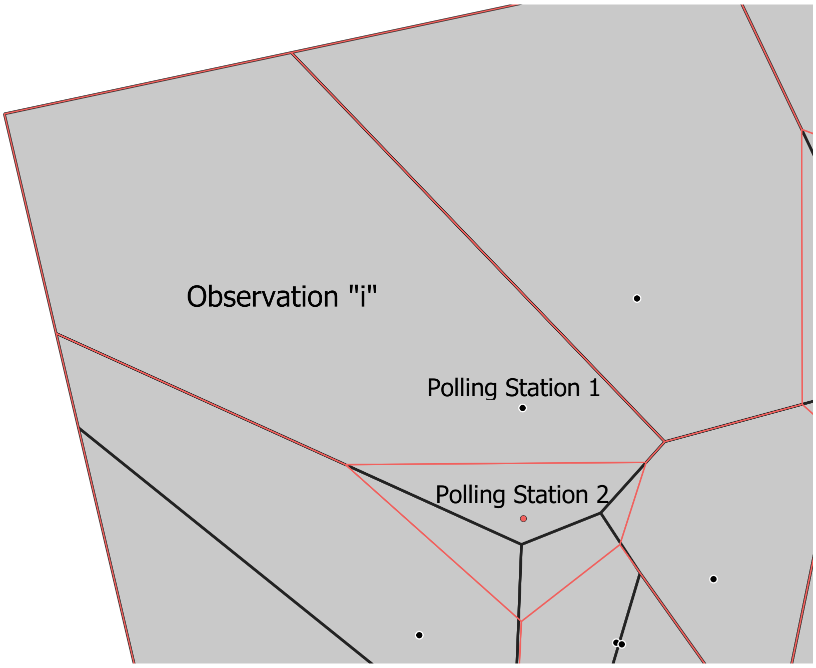


Figure 42 describes an example of how the weighting strategy works. Take observation "i" (the striped polygon). In 2006, its votes were entirely made out of Polling Station "1". Polling Station "2" was opened in 2010, which shrank Polling Station 1 Voronoi borders. Now, 100% of Polling Station "1" Voronoi Polygon and 50% of Polling Station "2" lie within observation "i".

Figure 42: Example of Weighting to get 2010 Political Outcome

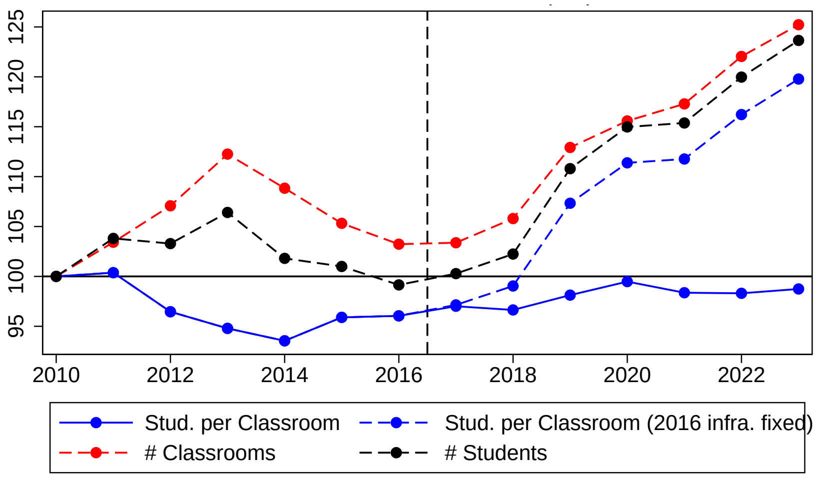


The number of votes a certain candidate "13" had in 2010 for observation "i" equals 100% the number of votes for "13" at polling station "1" summed with 50% polling station "2" votes for "13". Using the same strategy for the total number of votes, I will get the observation "i" share of votes for a candidate "13" in 2010. Treated Polygons will be the ones for which their center is less than one kilometer away from the closest shelter.

G Public Education

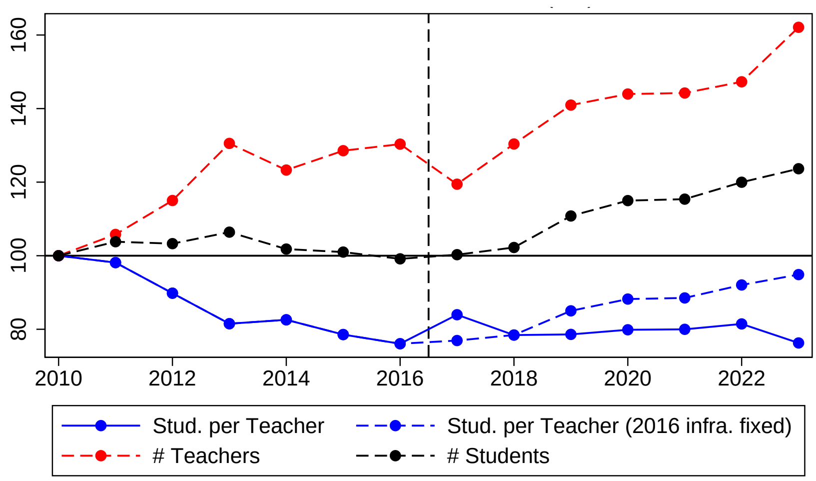
Aggregating the School Census yearly data at the municipal level allows us to examine how Boa Vista's public education system, as a whole, managed the influx of Venezuelan students. Both the number of classrooms and the number of teachers increased, helping keep classroom sizes and student-to-teacher ratios at levels similar to those registered pre-migration (see figures below). However, we do observe that classrooms were created at the expense of the destruction of other infrastructures such as libraries, reading rooms, and computer and science labs.

Figure 43: Classroom Size (2010=100) - Boa Vista's Public Schools



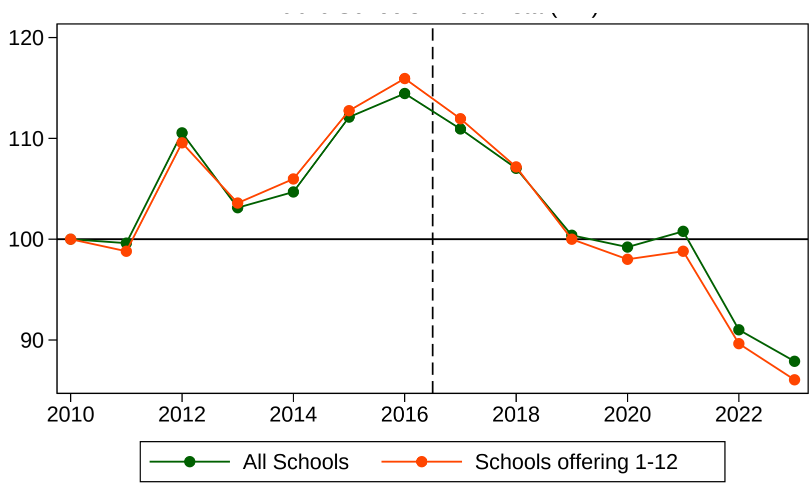
Notes: Boa Vista urban area public schools. *Source:* School Census (2010-2023).

Figure 44: Students per teacher ratio (2010=100) - Boa Vista's Public Schools



Notes: Boa Vista urban area public schools. *Source:* School Census (2010-2023).

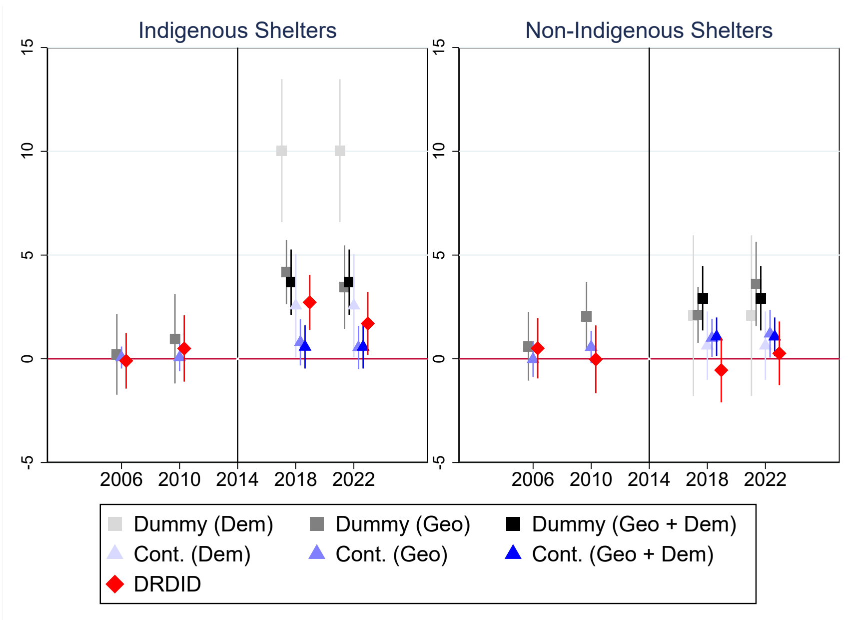
Figure 45: Number of libraries, reading rooms and labs (2010=100) - Boa Vista's Public Schools



Notes: Boa Vista urban area public schools. *Source:* School Census (2010-2023).

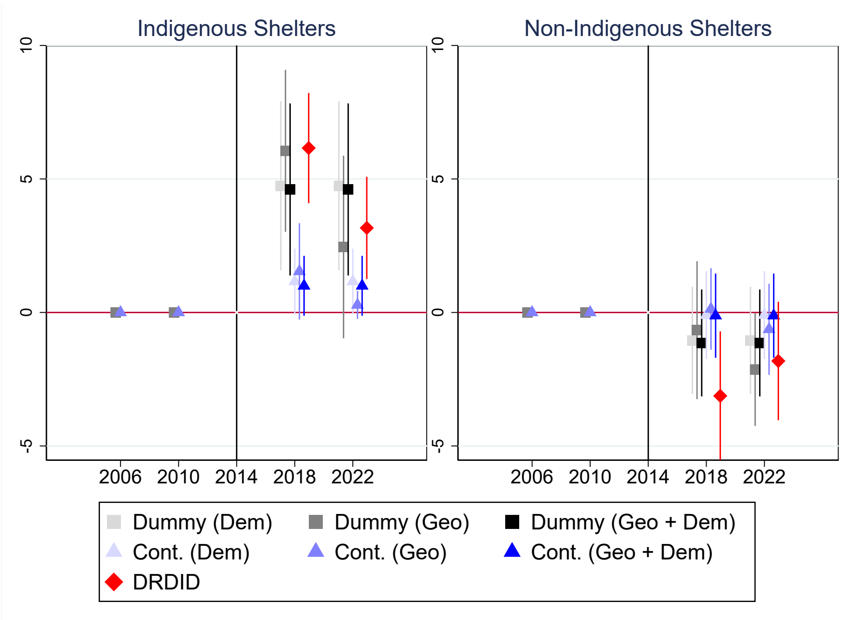
H Indigenous Vs Non-Indigenous Shelters Results

Figure 46: President Election (Challenger) Event Study - Indigenous Vs Non-Indigenous



Notes: 2014 is the baseline year. Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

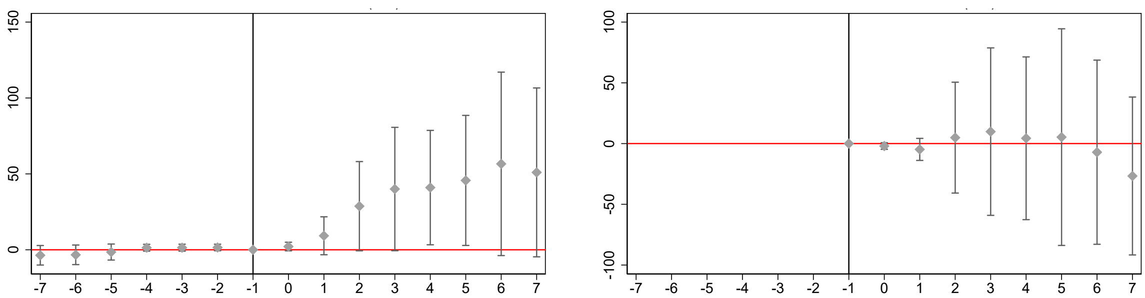
Figure 47: Governor Election (Far-Right) Event Study - Indigenous Vs Non-Indigenous



Notes: 2014 is the baseline year. PSL did not participate or support any candidate in the 2006 and 2010 state elections. Dependent Variable = % of valid votes. *Dem* = 23 demographic (age, education, and gender) controls; *Geo* = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

H.1 Public Education

Figure 48: Shelters' Effect on **Schools' Composition - Number of Venezuelan Students**

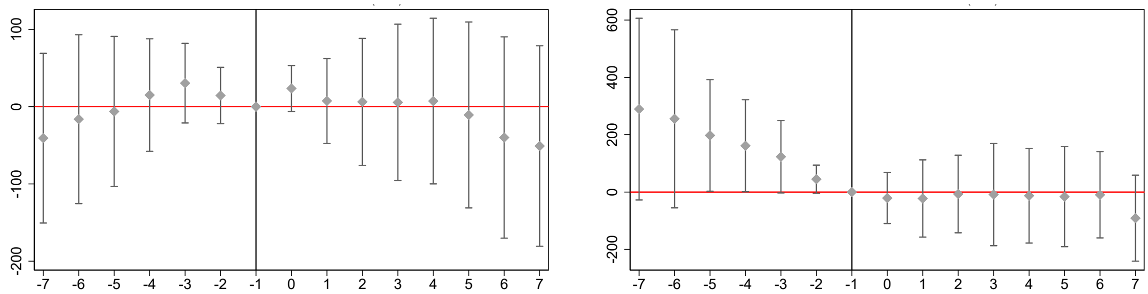


(a) Non-Indigenous Shelters

(b) Indigenous Shelters

Notes: Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from an Indigenous shelter. 2016 is the baseline year. The sample consists of all public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

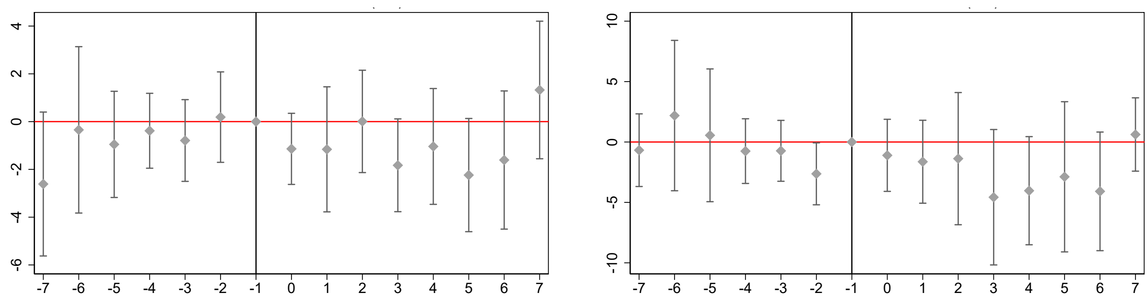
Figure 49: Shelters' Effect on **Schools' Composition - Number of Brazilian Students**



(a) Non-Indigenous Shelters (b) Indigenous Shelters

Notes: Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from an Indigenous shelter. 2016 is the baseline year. The sample consists of all public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

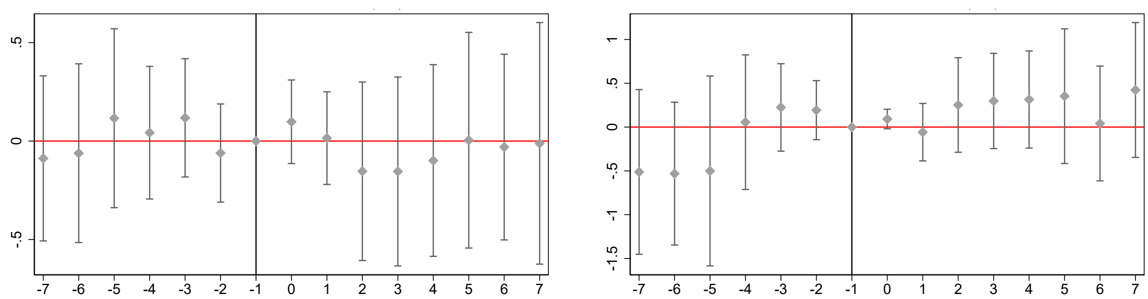
Figure 50: Shelters' Effect on **Schools' Congestion - Students per Teacher**



(a) Non-Indigenous Shelters (b) Indigenous Shelters

Notes: Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from an Indigenous shelter. 2016 is the baseline year. The sample consists of all public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

Figure 51: Shelters' Effect on Schools' **Infrastructure (number of reading rooms, libraries, labs)**

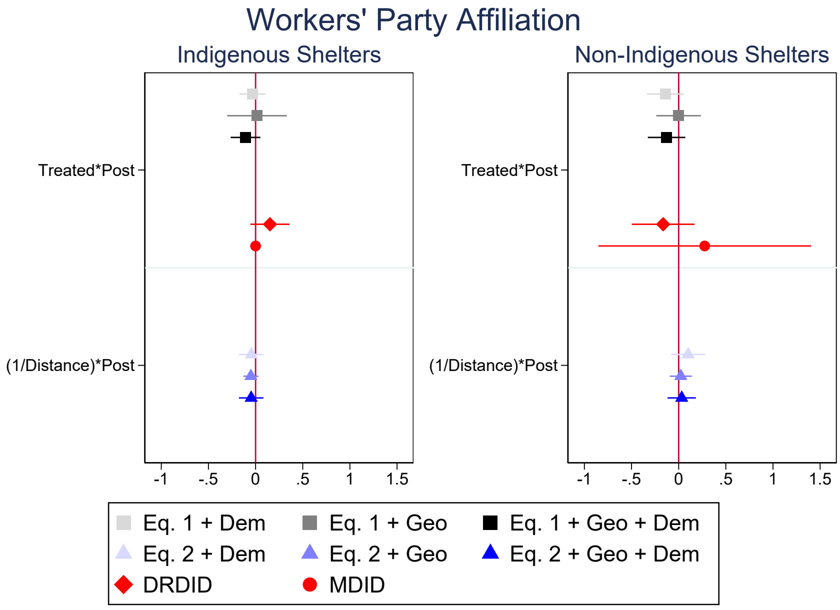


(a) Non-Indigenous Shelters (b) Indigenous Shelters

Notes: Event Study based on a "treatment" dummy equal to 1 for a school less than 1 km away from an Indigenous shelter. 2016 is the baseline year. The sample consists of all public schools in Boa Vista, including kindergarten, nursery schools, and grades 1 to 12.

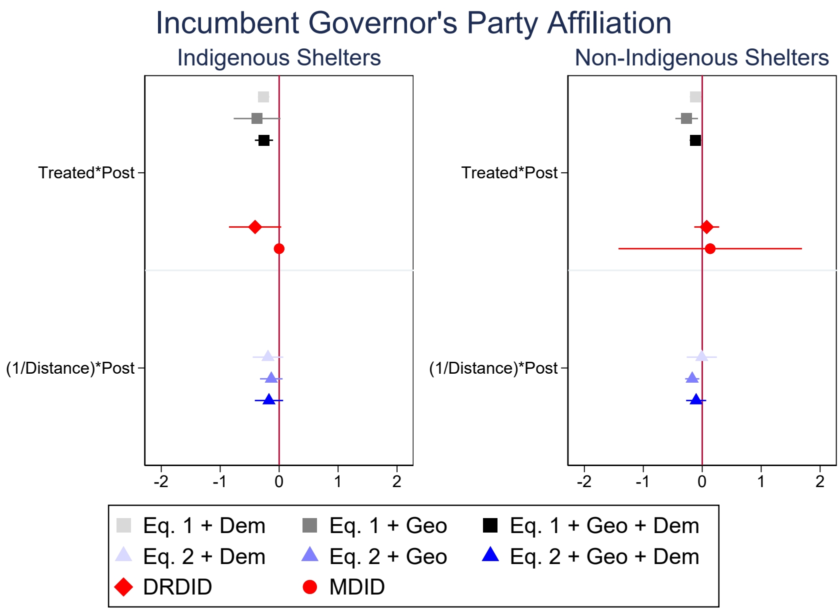
H.2 Party Affiliation

Figure 52: Party Affiliation Results



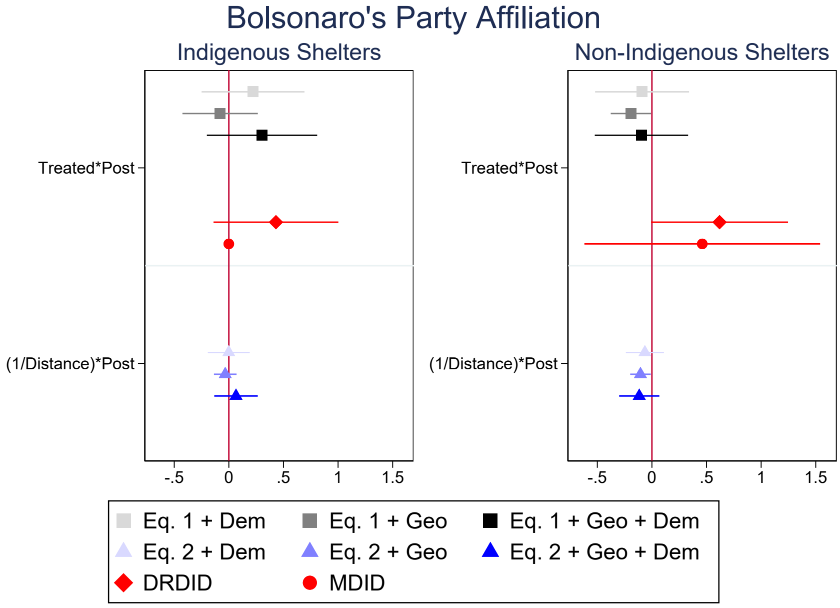
Notes: Dependent Variable = ihs transformation of the number of affiliates. Dem = 23 demographic (age, education, and gender) controls; Geo = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

Figure 53: Party Affiliation Results



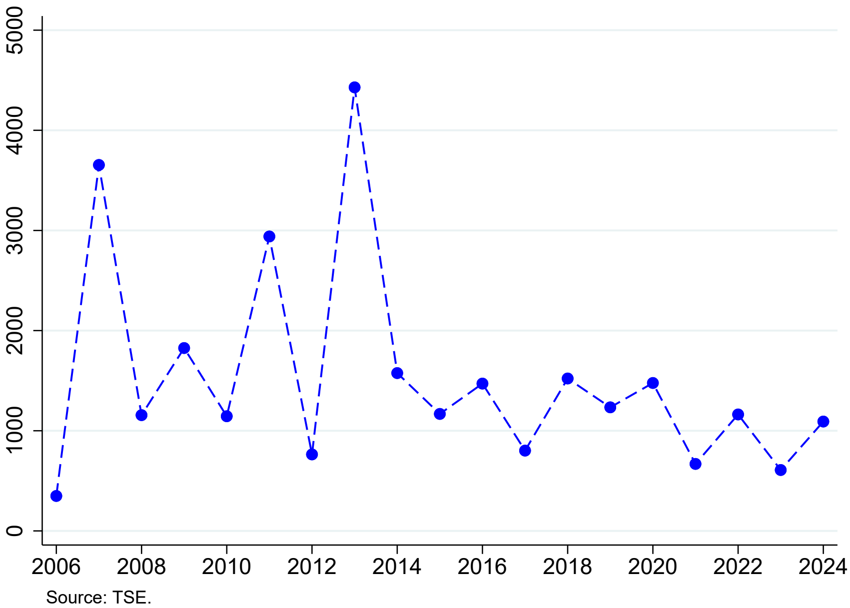
Notes: Dependent Variable = ihs transformation of the number of affiliates. Dem = 23 demographic (age, education, and gender) controls; Geo = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

Figure 54: Party Affiliation Results



Notes: Dependent Variable = ihs transformation of the number of affiliates. Dem = 23 demographic (age, education, and gender) controls; Geo = time dummies interacted with polling station distance to downtown. DRDID = Doubly Robust DiD. MDID = Matching DiD.

Figure 55: Number of New Affiliates (All Parties) - Boa Vista

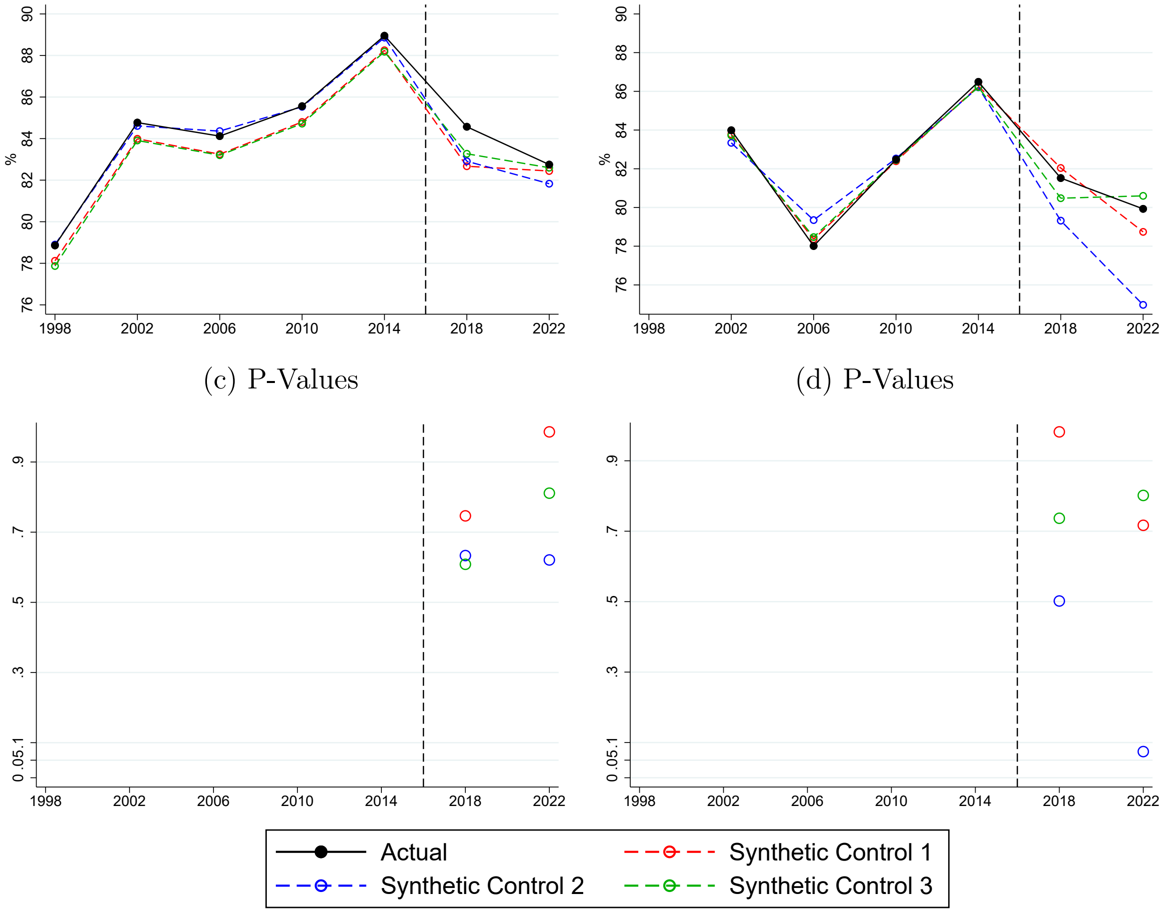


I Synthetic Control

Figure 56: Overall Effect of Reception Policy

(a) Turnout (1st Round) (b) Turnout (2nd Round)

(c) P-Values (d) P-Values



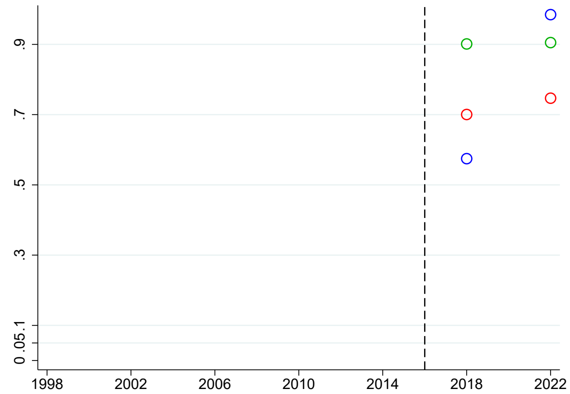
Notes: Synthetic Control 1: random 5% sample of Brazilian municipalities outside Roraima. *Synthetic Control 2*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average population to Boa Vista. *Synthetic Control 3*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average GDP per capita to Boa Vista.

Figure 57: Overall Effect of Reception Policy

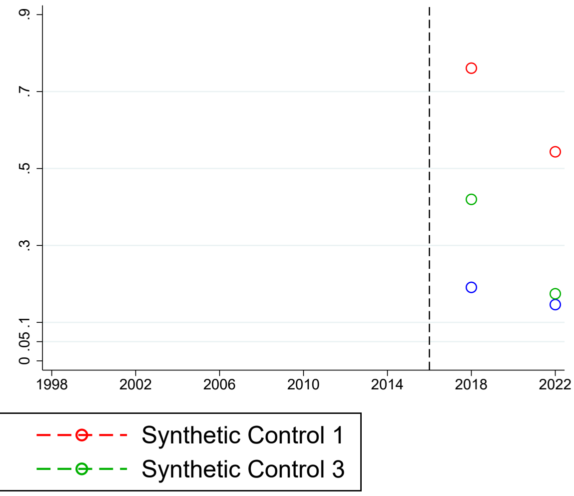
(a) Non-Valid for President (1st Round)

(b) Non-Valid for President (2nd Round)

(c) P-Values



(d) P-Values

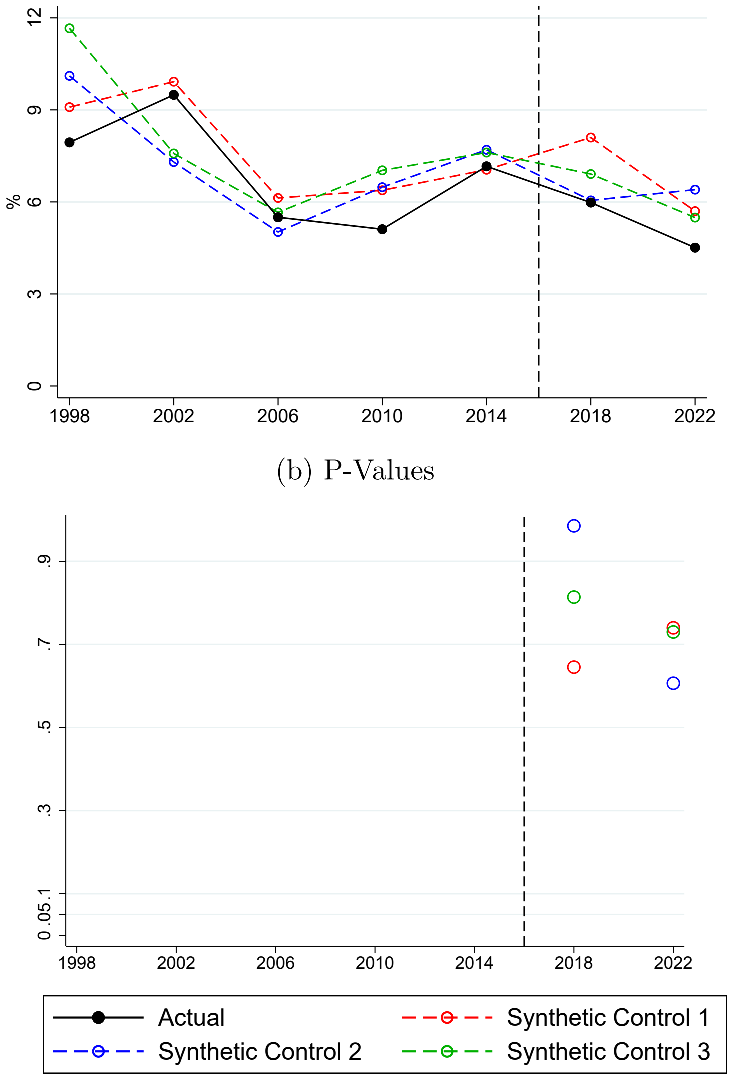


Notes: Synthetic Control 1: random 5% sample of Brazilian municipalities outside Roraima. *Synthetic Control 2*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average population to Boa Vista. *Synthetic Control 3*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average GDP per capita to Boa Vista.

Figure 58: Overall Effect of Reception Policy

(a) Non-Valid for Governor (1st Round)

(b) P-Values



Notes: Synthetic Control 1: random 5% sample of Brazilian municipalities outside Roraima. *Synthetic Control 2*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average population to Boa Vista. *Synthetic Control 3*: 5% sample of Brazilian municipalities outside Roraima with the closest pre-migration average GDP per capita to Boa Vista.

J Robustness

Figure 59: Random Inference - Y = Vote Share for Far-Right Governor

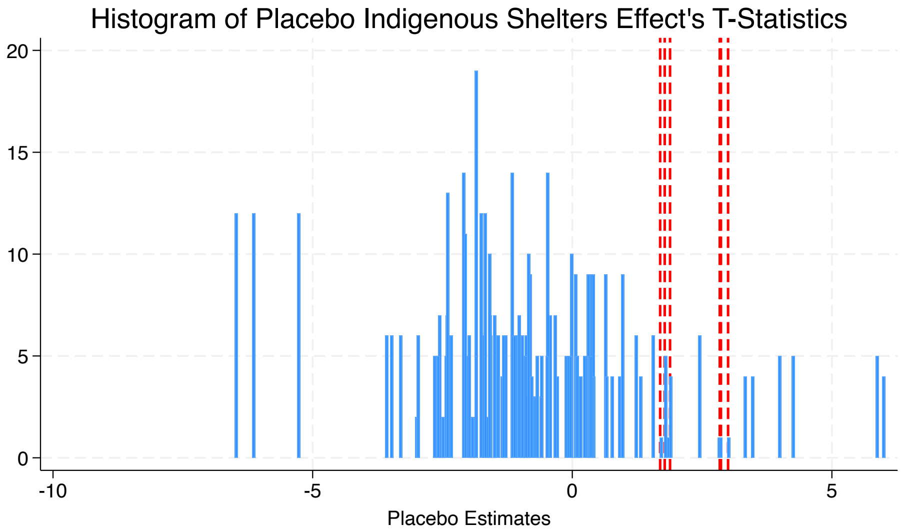


Figure 60: Random Inference - Y = Vote Share for Incumbent Governor

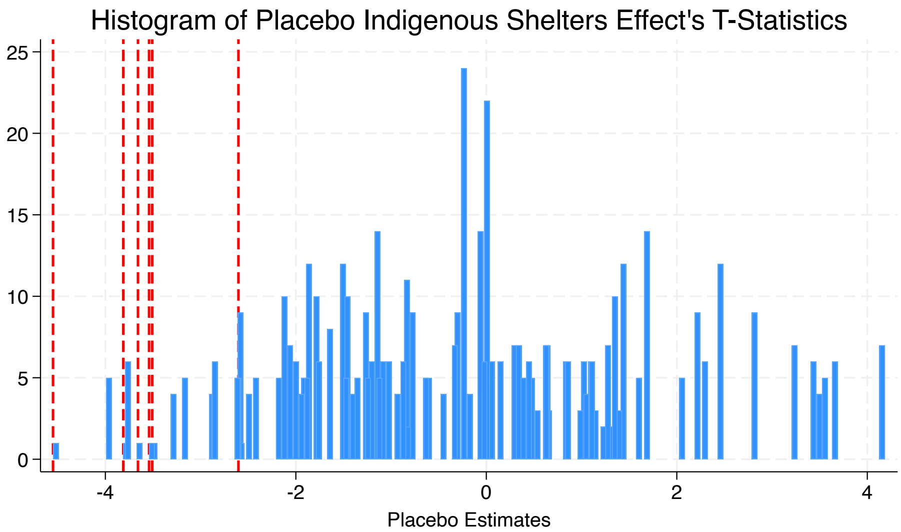


Figure 61: Random Inference - Y = Vote Share for Far-Right President (2nd Round)

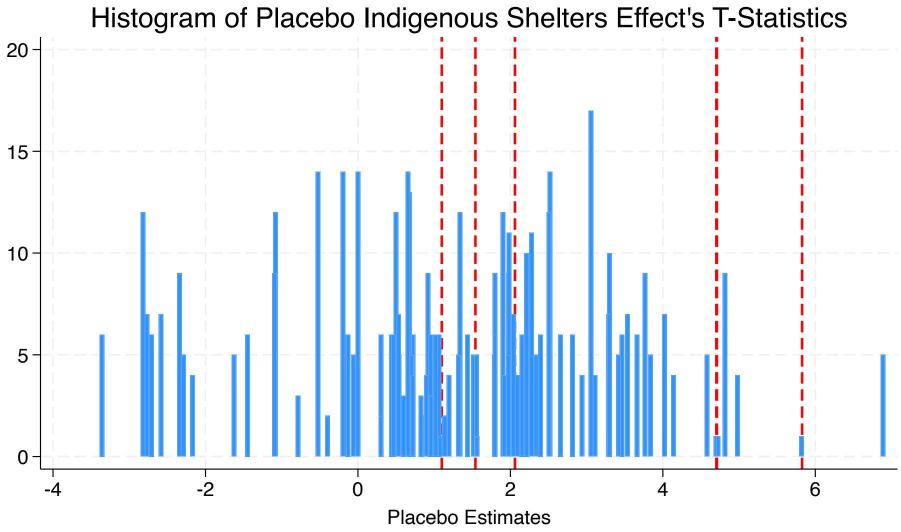


Table 22: Excluding polling stations treated by both types of shelters

	(1)	(2)	(3)	(4)	(5)	(6)
	Y = Vote Share Incumbent Governor					
Treated (Ind. Shelter)$_j$ × Post$_t$	-8.372***	-3.813***	-4.307**			
	(2.196)	(1.087)	(1.655)			
Treated (Non-Ind. Shelter)$_j$ × Post$_t$	0.588	-1.359	-0.398			
	(1.842)	(1.015)	(1.082)			
(1/ Dist. Ind. Shelter)$_j$ × Post$_t$				-2.532***	-1.401***	-1.200***
				(0.692)	(0.308)	(0.339)
(1/ Dist. Non-Ind. Shelter)$_j$ × Post$_t$				0.142	-0.542	-0.302
				(1.088)	(0.433)	(0.547)
Observations	476	952	476	476	952	476
R^2	0.960	0.963	0.976	0.956	0.963	0.975
	Y = Vote Share Far-Right Governor					
Treated (Ind. Shelter)$_j$ × Post$_t$	4.742***	4.253***	4.607***			
	(1.581)	(1.497)	(1.609)			
Treated (Non-Ind. Shelter)$_j$ × Post$_t$	-1.046	-1.399	-1.141			
	(1.003)	(1.007)	(1.000)			
(1/ Dist. Ind. Shelter)$_j$ × Post$_t$				1.158*	0.910*	0.998*
				(0.615)	(0.538)	(0.560)
(1/ Dist. Non-Ind. Shelter)$_j$ × Post$_t$				-0.099	-0.249	-0.119
				(0.820)	(0.768)	(0.788)
Observations	714	714	714	714	714	714
R^2	0.778	0.797	0.806	0.768	0.789	0.797
Voters' Dem. Controls	YES	NO	YES	YES	NO	YES
Distance to Downtown × Time FE	NO	YES	YES	NO	YES	YES

Notes: Robust standard errors are reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

Figure 62: Voting Outcomes Per Treatment Status Over Time